

Logic Gate

2 types

1. Basic gate (OR, AND, NOT)

2. Complex gate (NOR, NAND, XOR, XNOR)

OR $\rightarrow$  $A+B$

$$0+0=0$$

$$0+1=1$$

$$1+0=1$$

$$1+1=1$$

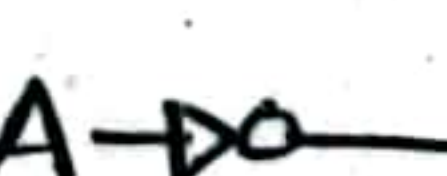
AND $\rightarrow$  AB

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

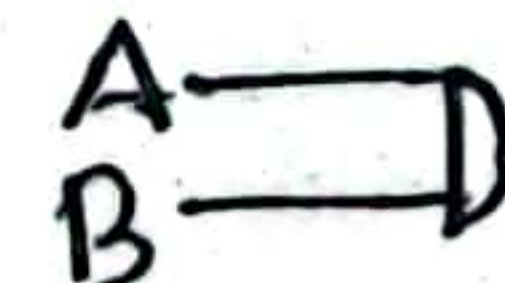
$$1 \cdot 1 = 1$$

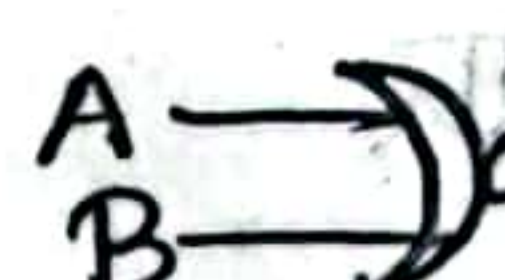
NOT $\rightarrow$  $\bar{A}$

$$\bar{\bar{A}} = A$$

$$\bar{0} = 1$$

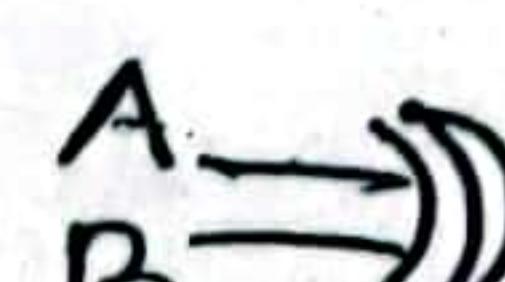
$$\bar{1} = 0$$

NAND $\rightarrow$  $\overline{AB}$

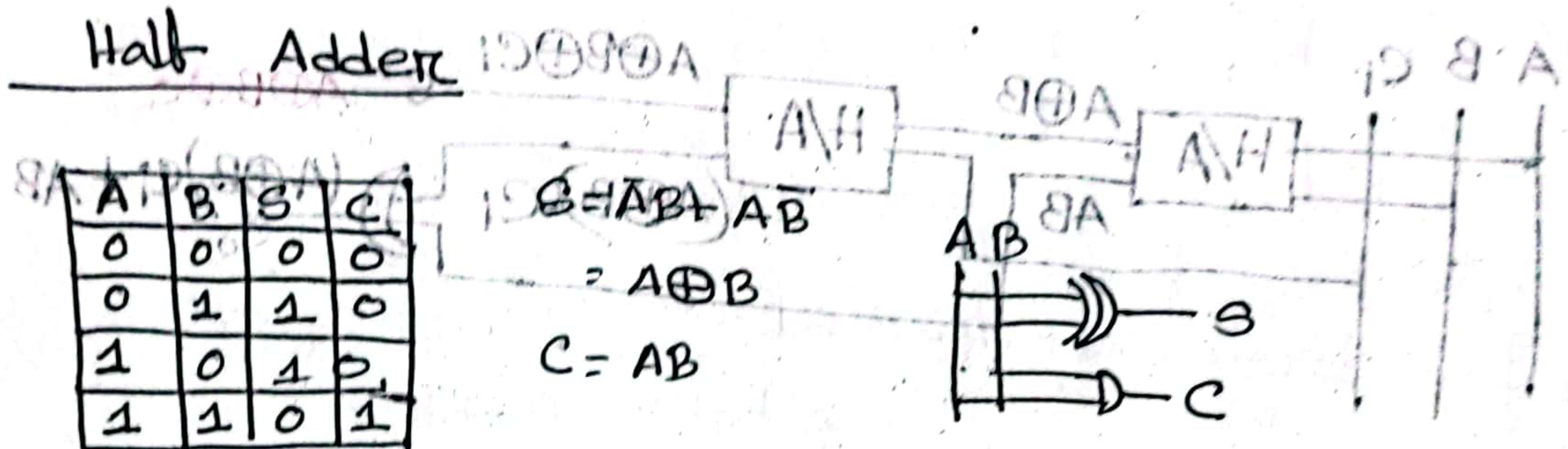
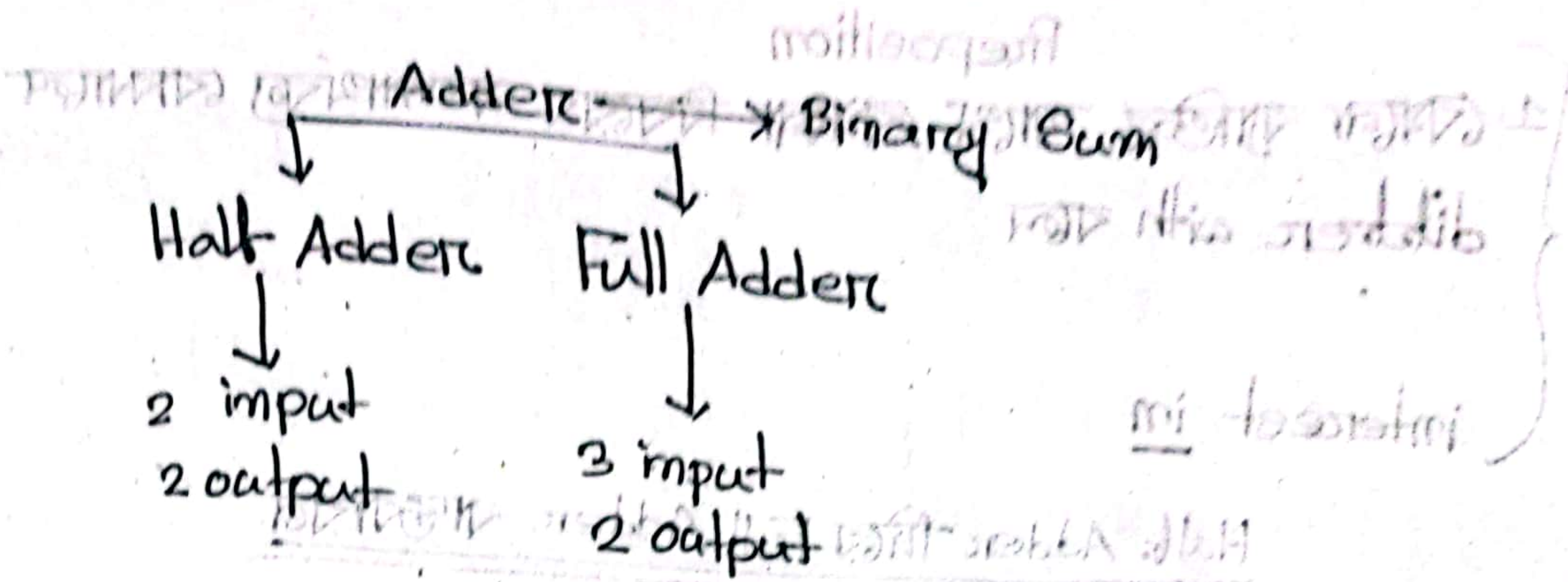
NOR $\rightarrow$  $\overline{A+B}$

XOR $\rightarrow$  $A \oplus B$
 $= \bar{A}B + A\bar{B}$

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

X-NOR $\rightarrow$  $\overline{A \oplus B}$
 $= \bar{A}\bar{B} + AB$

A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

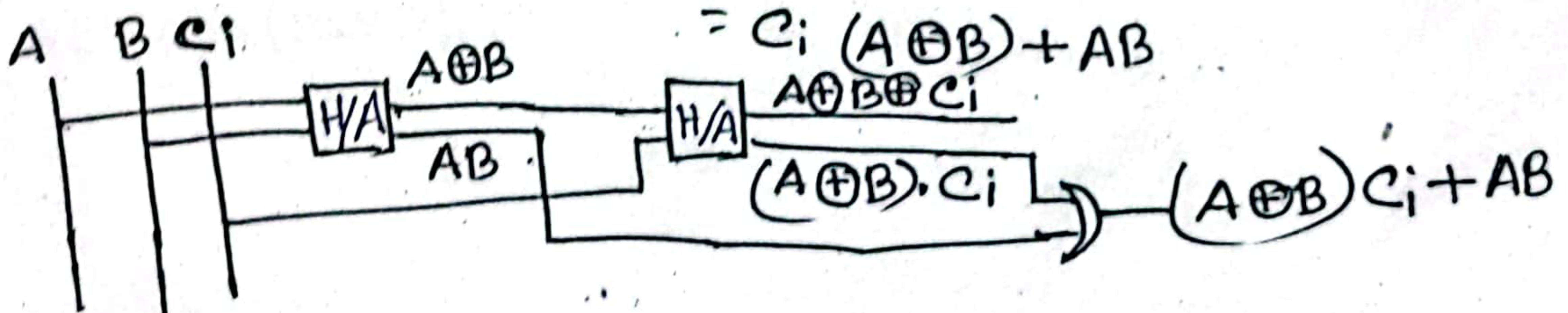


Half Adder দিয়ে Full Adder বাস্তবায়ন

Input			Output	
A	B	C _i	S	C _o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$\begin{aligned}
 S &= \overline{A}\overline{B}C_i + \overline{A}B\overline{C}_i + A\overline{B}\overline{C}_i + ABC_i \\
 &= \overline{A}(\overline{B}C_i + B\overline{C}_i) + A(\overline{B}\overline{C}_i + BC_i) \\
 &= \overline{A}(B \oplus C_i) + A(\overline{B \oplus C_i}) \\
 &= A \oplus B \oplus C_i
 \end{aligned}$$

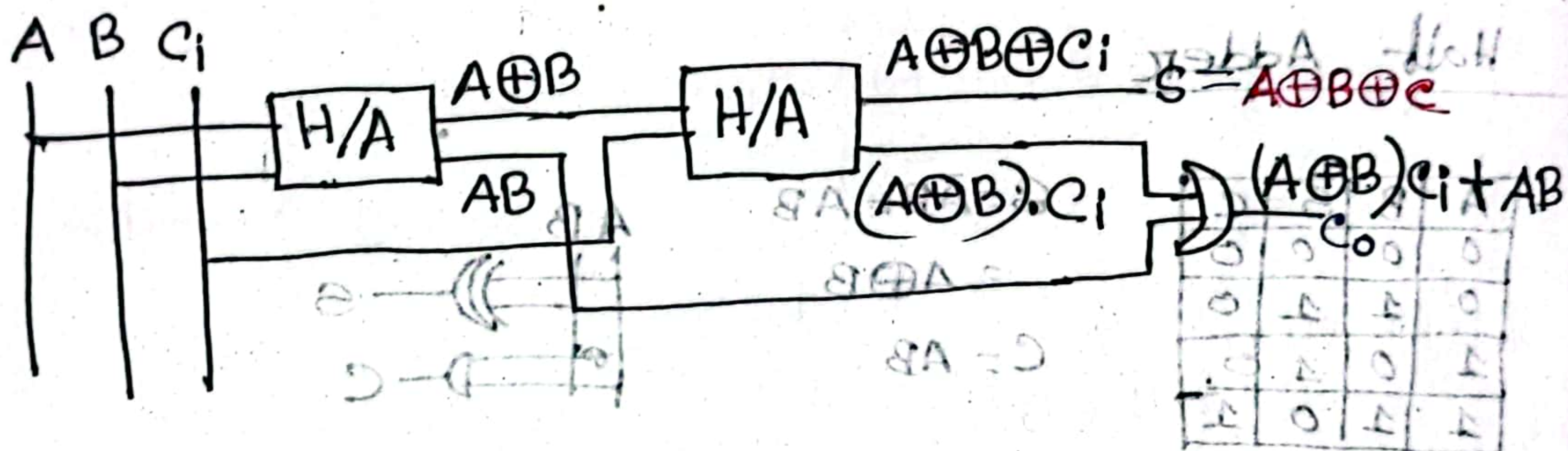
$$\begin{aligned}
 C_o &= \overline{A}BC_i + A\overline{B}C_i + AB\overline{C}_i + ABC_i \\
 &= C_i(\overline{A}B + A\overline{B}) + AB(\overline{C}_i + C_i) \\
 &= C_i(A \oplus B) + AB
 \end{aligned}$$



Reposition

- কোনো ব্যক্তির সাথে কোনো বিষয়ের মতপার্থক্য বোঝাতে
differ with বলে
interest in

Half Adder দিয়ে Full Adder বাস্তবায়ন



Half Adder দিয়ে Full Adder বাস্তবায়ন

$$AB + \bar{A}B + A\bar{B} + \bar{A}\bar{B} = 1$$

$$(AB + \bar{A}B) + (A\bar{B} + \bar{A}\bar{B}) =$$

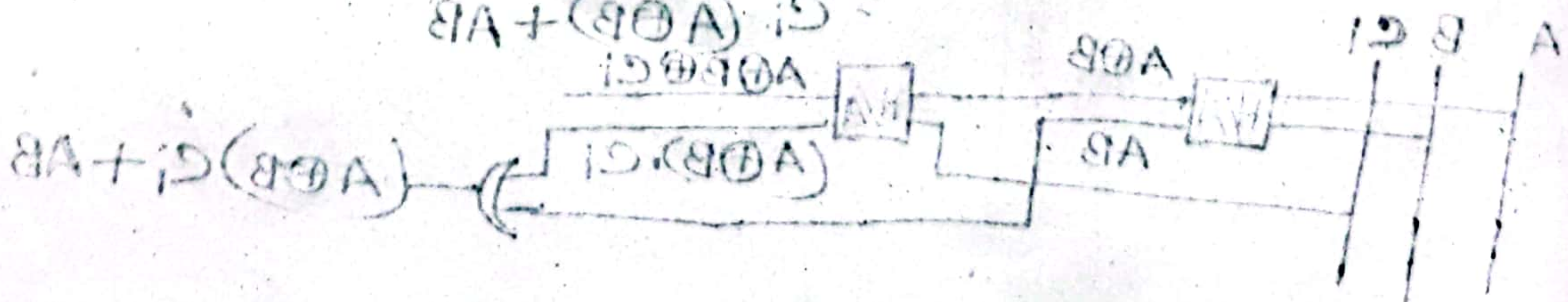
$$(B(A + \bar{A})) + (\bar{A}(B + \bar{B})) =$$

$$B(1) + \bar{A}(1) =$$

$$B + \bar{A} = 1$$

$$(B + \bar{A})(A + \bar{A}) =$$

$$BA + \bar{A}B + AB + \bar{A}\bar{B} =$$



$$(57)_{10} = (?)_2$$

$$= (?)_8$$

$$= (?)_{16}$$

$$\begin{array}{r} 2 \overline{) 57} \\ 28 - 1 \\ \hline 2 \overline{) 28} \\ 14 - 0 \\ \hline 2 \overline{) 14} \\ 7 - 0 \\ \hline 2 \overline{) 7} \\ 3 - 1 \\ \hline 2 \overline{) 3} \\ 1 - 1 \\ \hline 0 - 1 \end{array}$$

$$\begin{array}{r} 8 \overline{) 57} \\ 48 - 1 \\ \hline 0 - 7 \end{array}$$

$$\begin{array}{r} 16 \overline{) 57} \\ 32 - 2 \\ \hline 0 - 3 \end{array}$$

$$(57)_{10} = (111001)_2 \quad (57)_{10} = (71)_8 \quad (57)_{10} = (39)_{16}$$

$$(10101101)_2 = (?)_{10} \quad (?)_8 \quad (?)_{16}$$

$$\begin{aligned} (10101101)_2 &= 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ &= 128 + 32 + 8 + 4 + 1 \\ &= (173)_{10} \end{aligned}$$

$$\begin{array}{r} 010101101 \\ \underline{2 \quad 5 \quad 5} \end{array}$$

$$(10101101)_2 = (255)_8$$

$$(A53)_{16} = (?)_8$$

$$\begin{array}{r} 10101101 \\ \downarrow \quad \downarrow \\ 10 \quad 13 \\ \downarrow \quad \downarrow \\ A \quad D \end{array}$$

$$(10101101)_2 = (AD)_{16}$$

$$\begin{array}{r} 101001010011 \\ \underline{5 \quad 1 \quad 2 \quad 3} \end{array}$$

$$(A53)_{16} = (5123)_8$$

21-08-2023

স্মিগ-স্মিগ:

মিক্রোপ্রসেসর অডিও আর্কিট দ্বারা গঠিত একটি
বাইটবল মানিটাইনিং আর্কিট

এর ২ টি control Input 0 → low signal
২ টি control Output 1 → high signal

১ টি Flip-Flop → ১ টি Binary bit ধারণ করতে পারে

4 types

মানে বাথতে পারে

১. SR → Set-Reset

২. D

৩. JK

৪. T

* ১ byte এর Shift-Register তৈরি করতে
হলে ৮ bit লাগে

IC → ইন্টিগ্রেটেড আর্কিট

1 KB = 1024 byte

Uses of Flip-Flop

১. রেজিস্টার তৈরি
২. কাউন্টার তৈরি
৩. মেমোরির উপাদান তৈরি

MSI → Medium Scale Integration

LSI → Large Scale Integration

VLSI → Very Large Scale Integration

Base দিয়ে গুল-ভাগ করা হচ্ছে (যে কোনো অঙ্ক পাওয়া গেল)

$$(0.75)_{10} = (?)_2 \quad (?)_8 \quad (?)_{16}$$

$$0.75 \times 2 = 1$$

$$0.5 \times 2 = 1$$

$$0.75 \times 8 = 6$$

$$0.75 \times 16 = 12$$

ନିମ୍ନଲିଖିତ ସମସ୍ୟା

$$(1011.1101)_2 = (?)_{10}$$

$$A + \bar{A} = 1$$

$$= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + \frac{1}{2} + \frac{1}{4} + \frac{1}{16}$$

$$= 8 + 2 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{16}$$

$$(32.6)_8 = (?)_{10}$$

$$= 3 \times 8^1 + 2 \times 8^0 + 6 \times 8^{-1}$$

$$= 24 + 2 + \frac{6}{8}$$

Associative law

$$A + (B + C) = (A + B) + C$$

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

ନିମ୍ନଲିଖିତ ସମସ୍ୟା

$$0 = 0 \cdot A$$

$$A = 0 + A$$

$$A = 1 \cdot A$$

$$1 = 1 + A$$

$$A = A \cdot A$$

$$A = A + A$$

$$0 = A \cdot A$$

$$1 = A + A$$

ନିମ୍ନଲିଖିତ ସମସ୍ୟା

$$A + B + A$$

$$AB = BA$$

Distributive law

$$A(B + C) = AB + AC$$

$$A + BC = (A + B)(A + C)$$

$$A + \bar{A}B = (A + \bar{A})(A + B)$$

$$= A + B$$

$$A \cdot (B + C) = AB + AC$$

NOT

AND OR

The apex of a triangle — Top

উপপাদ্য

মৌলিক উপপাদ্য

$$A+0=A$$

$$A \cdot 0=0$$

$$A+1=1$$

$$A \cdot 1=A$$

$$A+A=A$$

$$A \cdot A=A$$

$$A+\bar{A}=1$$

$$A \cdot \bar{A}=0$$

বিনিময় উপপাদ্য

$$A+B=B+A$$

$$AB=BA$$

Distributive Law

$$A(B+C)=AB+AC$$

$$A+BC=(A+B)(A+C)$$

$$A+\bar{A}B=(A+\bar{A})(A+B) \\ =A+B$$

$$A \cdot (B+C)=AB+AC$$

Priority

NOT > AND > OR

ডি. মরগানের উপপাদ্য

$$\overline{A+B} = \bar{A} \cdot \bar{B}$$

$$\overline{AB} = \bar{A} + \bar{B}$$

Associative law

$$A+(B+C)=(A+B)+C$$

$$A \cdot (BC)=(AB) \cdot C$$

বুলিয়ান অ্যালজেব্রা

→ বীজগণিত

বিজ্ঞানী ডাড্‌ বুল প্রদত্ত গণিত

1 → High/True

0 → Low/False

Redundancy Theorem

$$AC + \bar{A}\bar{B} + BC$$

→ # 3 টি টার্ম থাকতে হবে

প্রত্যেক টার্ম এ 2 টি করে টার্ম থাকতে হবে।

← বাদ যাবে

$$= \bar{A}\bar{B} + BC$$

একটি টার্মের complement থাকবে।

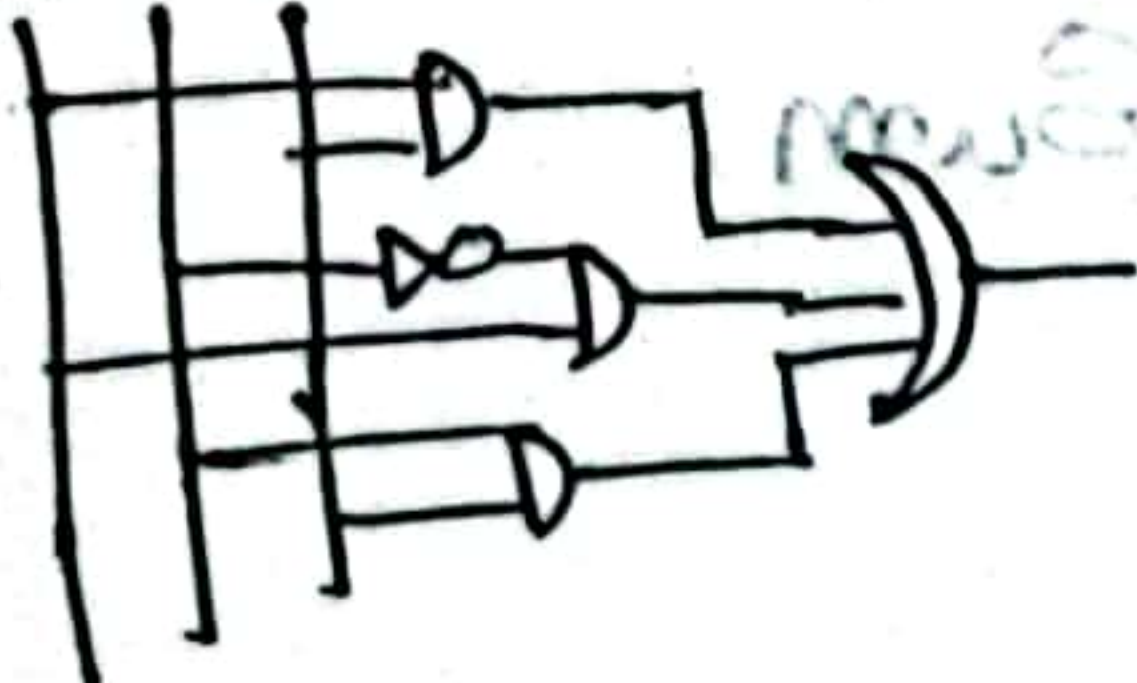
$$\begin{aligned} A + AB \\ &= A(1 + B) \\ &= A \end{aligned}$$

যে variable এর complement থাকবে, মেজের complement ছাড়া term দিতে হবে, / থাকবে। অন্য term-টি বাদ যাবে।

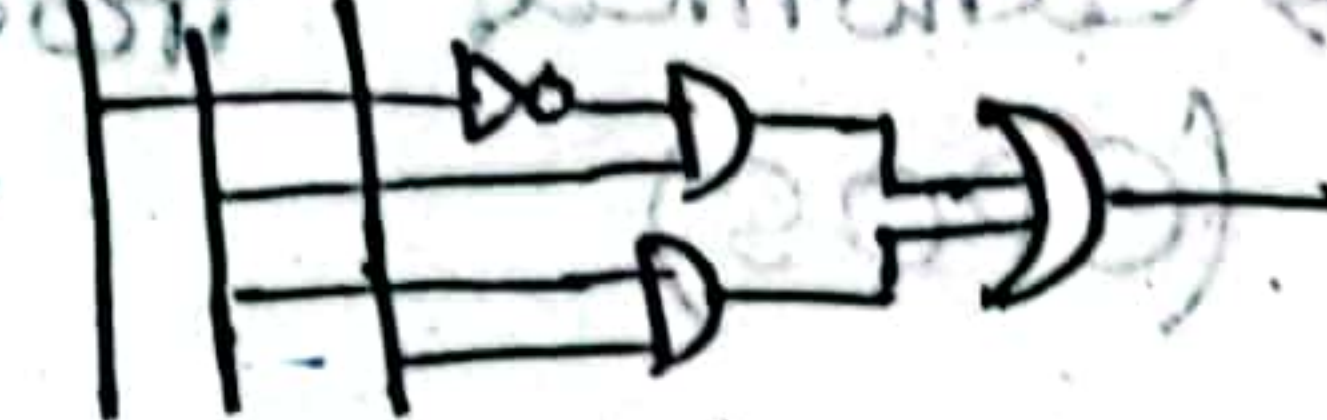
$$(1 + B + A)(1 + B + A)(1 + B + A) = (1 + B + A)$$

$$AC + \bar{A}\bar{B} + BC$$

ABC



ABC



$$AB' + BC + AC$$

$$= AB' + BC$$

→ বাদ যাবে

Canonical form

যে সুনির্দিষ্ট এক্সপ্রেসনের প্রত্যেকটা টার্ম এ অবশ্যই চমক থাকবে মোট Canonical form.

Complement/True
যে কোনো ভাবে
চমক থাকতে পারে

$$F(A, B, C) = A'BC + ABC$$

3 variable

→ 3 টি চমকই আছে

তাই এটি একটি Canonical form

Canonical Sum of Product
(CSOP)

$$F(A, B, C) = ABC' + A'B'C' + AB'C + ABC$$

$$F(A, B, C, D) = ABC'D + AB'C'D + A'B'C'D + ABCD$$

$$F(A, B, C) = (A+B'+C)(A'+B'+C)(A+B+C)$$

→ প্রত্যেক Term এই 3 টি করে চমক আছে.

→ Canonical Product of Sum
(CPOS)

Standard term/
Minimal term

$$B + A = (B + A) = 1$$

প্রত্যেক Term এ অবশ্যই চলক থাকবে না। কোনো না কোনো চলক Missing থাকবে।

$$F(A, B, C) = A + BC$$

এখানে A missing আছে।
 → এখানে শুধু A আছে। B ও C missing আছে।
 → এটি একটি Standard term

Standard/Minimal term

Standard/Minimal Sum of Product

$$F(A, B, C) = A'BC + AB'C + AB + C$$

→ Standard Sum of Product

$$F(A, B, C) = (A + B) \cdot (\bar{A} + B + C) (A + B + C')$$

→ Standard Product of Sum

Minterm



গুণ



High signal নিয়ে কাজ করে

$(m_0 - m_7)$ 3 variable

Maxterm



যোগ



Low signal নিয়ে কাজ করে

$(M_0 - M_7)$ 3 variable

Input			Minterms		Maxterms	
x	y	z	Terms		Terms	
0	0	0	$x'y'z'$	m_0	$x+y+z$	M_0
0	0	1	$x'y'z$	m_1	$x+y+z'$	M_1
0	1	0	$x'yz'$	m_2	$x+y'+z$	M_2
0	1	1	$x'yz$	m_3	$x+y'+z'$	M_3
1	0	0	$xy'z'$	m_4	$x'+y+z$	M_4
1	0	1	$xy'z$	m_5	$x'+y+z'$	M_5
1	1	0	xyz'	m_6	$x'+y'+z$	M_6
1	1	1	xyz	m_7	$x'+y'+z'$	M_7

{ m বা M এর subscript value হবে input গুলোর
Decimal value

1 0 1 ——— 4 2 1
 1 0 1
 ↪ M_5 / m_5

SOP
 $F = A$

Minimal to Canonical form

$$= A(B+B')$$

$$= AB + AB'$$

$$= AB(C+C') + AB'(C+C')$$

$$= ABC + AB'C' + AB'C + AB'C'$$

$$= m_7 + m_6 + m_5 + m_4$$

$$F = A + B'C$$

$$= A(B+B') + B'C(A+A')$$

$$= AB + AB' + AB'C + A'B'C$$

$$= AB(C+C') + AB'(C+C') + AB'C + A'B'C$$

$$= ABC + ABC' + AB'C + AB'C' + AB'C + A'B'C = (ABC + ABC' + AB'C + AB'C' + A'B'C)$$

$$= ABC + ABC' + AB'C + AB'C' + A'B'C$$

$$= m_7 + m_6 + m_4 + m_5 + m_1$$

Sum of Minterms/Sum of Product (SOP)

Product of Sum/Product of Maxterms (POS)

SOP → কয়েকটি ডিজনিয়ের গুণফলকে যদি যোগ আকারে প্রকাশ করি
 তাহলে যেটা SOP/SOMinterm

POS → কয়েকটি চর্মকের যোগফলকে যদি গুণ আকারে প্রকাশ করি
 তাহলে যেটা POS/POMaxterm

$$M = m'$$

SOP ($\sum$ চিহ্ন)

$$F(A, B, C) = \sum m(1, 4, 5, 6, 7)$$

$$= m_1 + m_4 + m_5 + m_6 + m_7$$

$\sum \rightarrow$ চিহ্ন দ্বারা বোঝায় কয়েকটি টার্মের গুণফলের যোগ (Sum of Product)

1 দ্বারা $\rightarrow$ মিনটার্ম m_1

4 $\rightarrow m_4$

5 $\rightarrow m_5$

6 $\rightarrow m_6$

7 $\rightarrow m_7$

$$F(A, B, C) = \sum m(1, 4, 5, 6, 7)$$

$$= \sum m_1 + m_4 + m_5 + m_6 + m_7$$

$$F'(A, B, C) = \sum (0, 2, 3) = (m_0 + m_2 + m_3)'$$

$$= M_0 \cdot M_2 \cdot M_3 = \prod (0, 2, 3)$$

POS ($\prod$ চিহ্ন)

$$F(A, B, C) = \sum (1, 4, 5, 6, 7)$$

এই function এর complement হবে

$$F'(A, B, C) = \sum (0, 2, 3)$$

$$= (m_0 + m_2 + m_3)'$$

$$= \prod (0, 2, 3)$$

$m' = M \rightarrow$ minterm এর complement হবে Maxterm

SOP থেকে POS এ convert

Minterm থেকে Maxterm-এ convert

⇒ Let,

$F(A, B, C) = \sum (1, 4, 5, 6, 7) \rightarrow$ এটি একটি SOP/EO minterm function.

এই function এর complement বের করলে মৌলিক হবে POS/PO Maxterm.

$$\begin{aligned} F'(A, B, C) &= \sum (0, 2, 3) \\ &= (m_0 + m_2 + m_3)' \\ &= m_0' m_2' m_3' \\ &= M_0 M_2 M_3 \\ &= \pi (0, 2, 3) \end{aligned}$$

মক্সটার্মের Location	B	A
$\overline{A+B} = \overline{A} \overline{B}$	0	0
$\overline{AB} = \overline{A} + \overline{B}$	0	1
	1	0
	1	1

কোনো function এর complement হবে, function এ যে কয়েকটা Minterm থাকবে না (Missing Minterm) সেগুলোর (Sum of Product)' বা Product of Sum

Example:

$$\begin{aligned} F(A, B, C) &= \sum (2, 5, 7) \\ &= m_2 + m_5 + m_7 \end{aligned}$$

$$\begin{aligned} F'(A, B, C) &= \sum (0, 1, 3, 4, 6) \\ &= (m_0 + m_1 + m_3 + m_4 + m_6)' \\ &= m_0' m_1' m_3' m_4' m_6' \\ &= M_0 M_1 M_3 M_4 M_6 \\ &= \pi (0, 1, 3, 4, 6) \end{aligned}$$

এখানে মক্সটার্মের লোকেশন বোঝানো হয়েছে।

Singular formula

agendum (আলোচনা)
oasis (মরুভূমি)

28-08-2023

Plural formulae

agenda
oases

Minterm
গুণ
AB

Maxterm
যোগ
A+B

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

	A	B	Y
0	0	0	0
1	0	1	1
2	1	0	0
3	1	1	1

$Y(A,B) = m_1 + m_3$
[SOP] = $\sum m(1,3)$

SOP (High)

0 $\rightarrow$ complement

$Y(A,B) = M_0 \cdot M_2$

[POS] = $\prod M(0,2)$

POS (Low)

0 $\rightarrow$ Normal term

SOP: $Y(A,B,C) = \sum m(0,2,3,6,7)$ $\rightarrow$ minterm (High signal)
 $= m_0 + m_2 + m_3 + m_6 + m_7$

POS: $Y(A,B,C) = \prod M(1,4,5)$ $\rightarrow$ Maxterm (low signal)
 $= M_1 \cdot M_4 \cdot M_5$

Canonical form

$Y(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C} + ABC$

$Y(A,B,C) = (A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+B+\bar{C})$

Minterm এর subscript হবে চমকগুলোর মানের যোগফলের Decimal value.

Simplification SOP

$$Y(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C} + ABC$$

$$= \bar{A}\bar{C}(\bar{B}+B) + \bar{A}BC + ABC\bar{C} + C$$

$$= \bar{A}\bar{C} + \bar{A}BC + AB$$

$$= \bar{A}(\bar{C} + BC) + AB$$

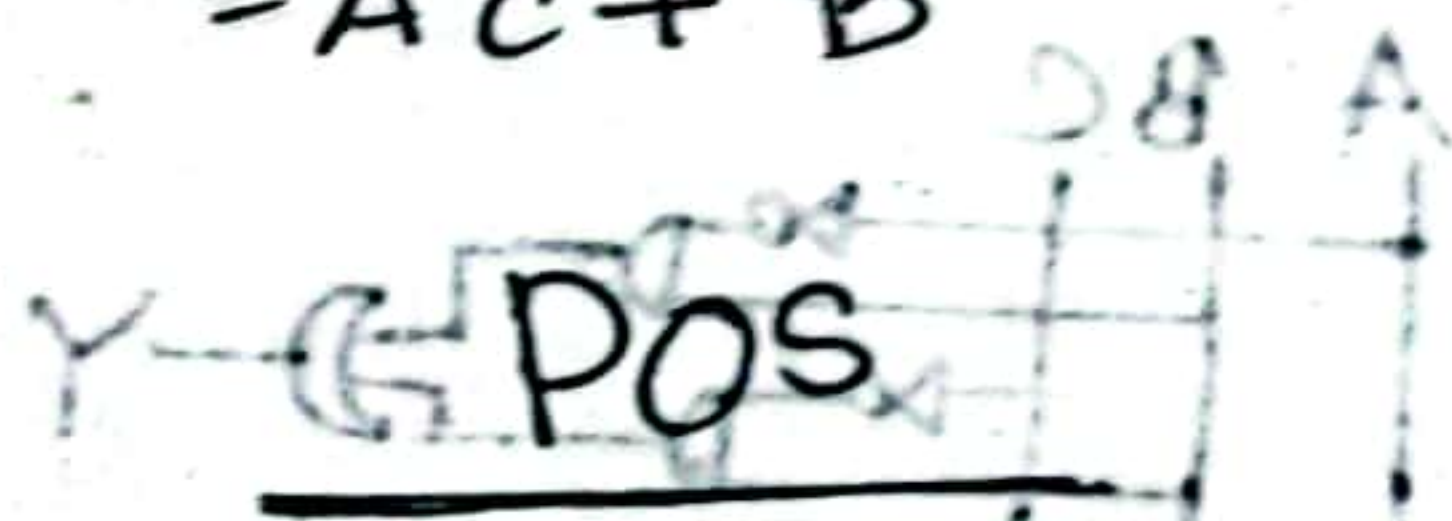
$$= \bar{A}(\bar{C} + B)(\bar{C} + C) + AB$$

$$= \bar{A}(\bar{C} + B) + AB$$

$$= \bar{A}\bar{C} + \bar{A}B + AB$$

$$= \bar{A}\bar{C} + B(\bar{A} + A)$$

$$= \bar{A}\bar{C} + B$$



$$Y(A, B, C) = (A+B+\bar{C}) \cdot (\bar{A}+B+C) (\bar{A}+B+\bar{C})$$

$$= (A+B+\bar{C}) (P+C) (P+\bar{C})$$

$$= (A+B+\bar{C}) \cdot P$$

$$= (A+B+\bar{C}) (\bar{A}+B)$$

$$= \bar{A}A + \bar{A}B + \bar{A}\bar{C} + AB + BB + B\bar{C}$$

$$= \bar{A}B + \bar{A}\bar{C} + AB + B + B\bar{C}$$

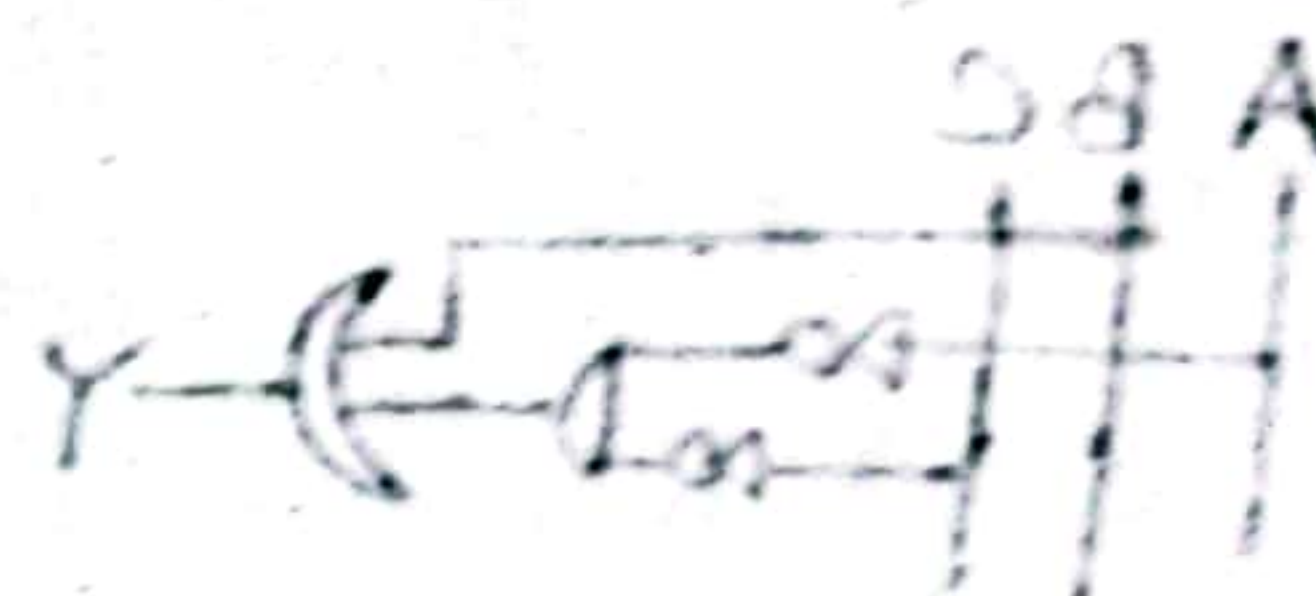
$$= \bar{A}B + \bar{A}\bar{C} + AB + B(1+\bar{C})$$

$$= \bar{A}\bar{C} + \bar{A}B + AB + B$$

Distribution Law

$$(A+BC) = (A+B)(A+C)$$

$$3A + 8A = Y$$



Let,

$$\bar{A}+B = P$$

$$(P+C)(P+\bar{C})$$

$$= PP + PC + P\bar{C} + C\bar{C}$$

$$= P + PC + P\bar{C}$$

$$= P + P(C+\bar{C})$$

$$= P + P$$

$$= P$$

$$= \bar{A}\bar{C} + \bar{A}B + B(A+1)$$

$$= \overline{A} \overline{C} + \overline{A} B + B$$

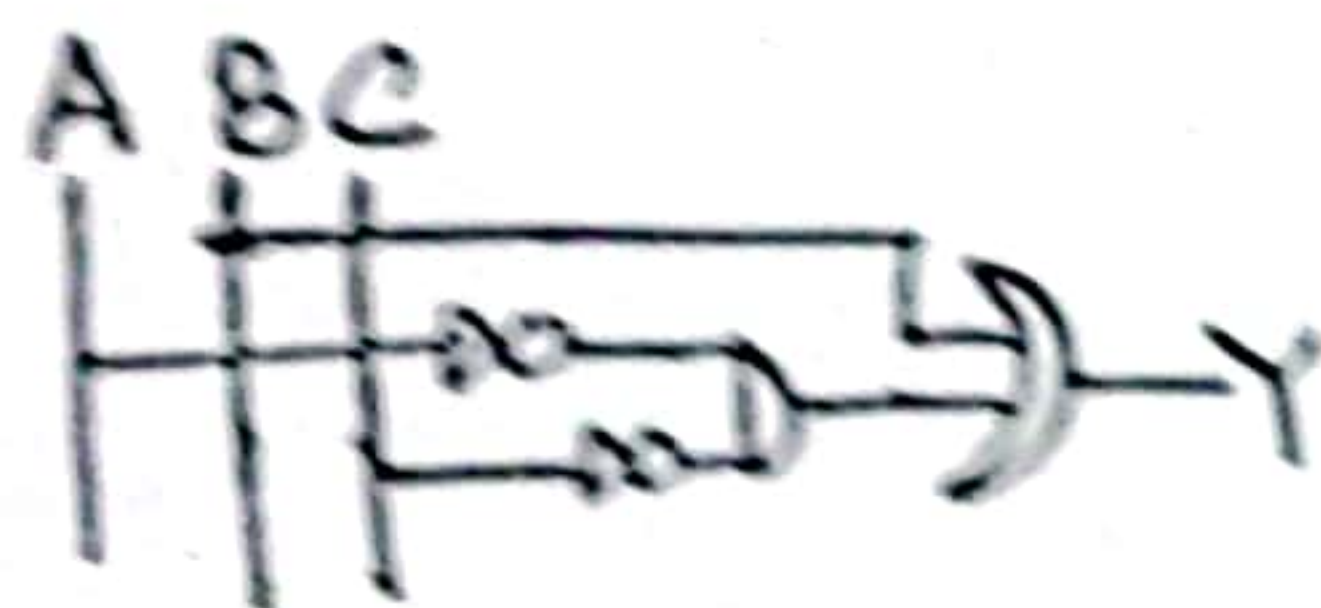
$$= \bar{A}\bar{C} + B(\bar{A}+1)$$

$$= \bar{A}\bar{C} + B$$

$$= (\bar{A} + B) = (\bar{C} + B)$$

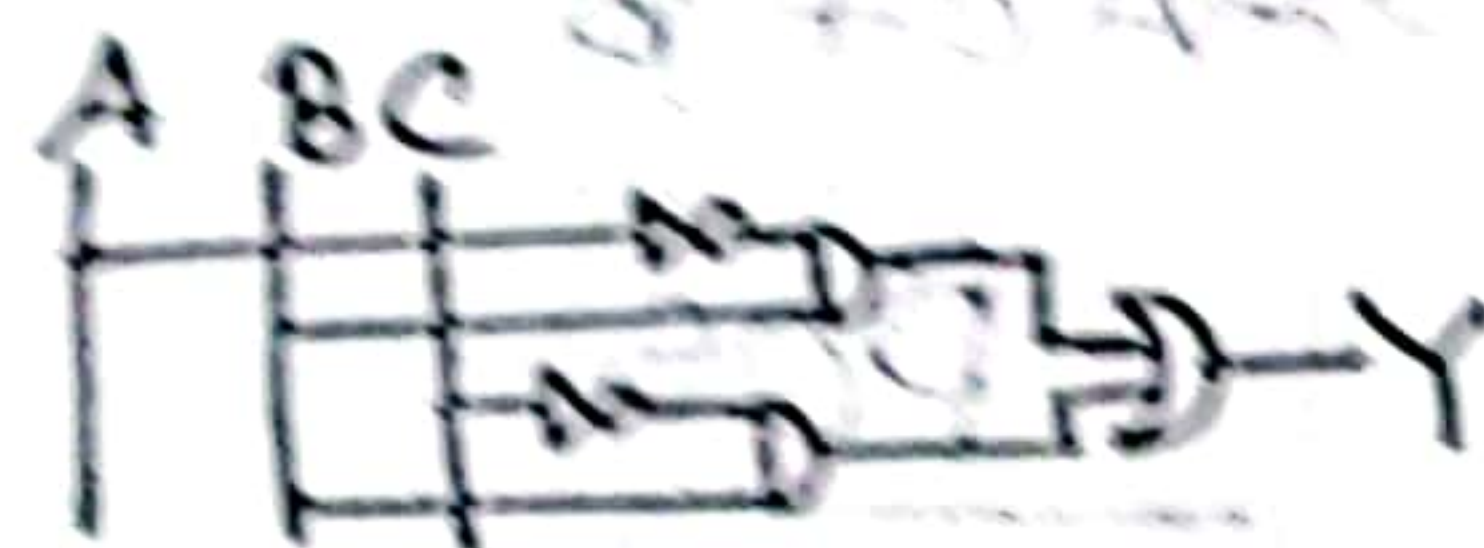
SOP

$$Y = B + \overline{A}C$$



POS

$$Y = (A+B)(\bar{C}+B)$$



K Map

$$F = \bar{A}B + \bar{A}\bar{B}$$

A \ B	0	1
0	1	1
1		

→ B ও B আছে
→ $\bar{A}$

$$F = \bar{A}$$

$$F = AB + \bar{A}B + A\bar{B}$$

A \ B	0	1
0		1
1	1	1

$$F = A + B$$

$$F = AB + \bar{A}B$$

A \ B	0	1
0		1
1		1

→ A ও $\bar{A}$ আছে
→ B

$$F = \bar{A}B\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}$$

A \ B \ C	00	01	11	10
0			1	1
1	1	1		

→ C, $\bar{C}$ আছে, বাদ

$\bar{A}B$ থাকবে শুধু

$$F = A\bar{B} + \bar{A}B$$

$$\bar{A} + \bar{C}A + \bar{C}\bar{A} + \bar{C}A = \bar{A}$$

A \ B \ C	00	01	10	11
0		1	1	
1				1

$$\begin{aligned} & AB + \bar{A}B + A\bar{B} \\ &= B(A + \bar{A}) + A\bar{B} \\ &= B + A\bar{B} \\ &= (B + \bar{B})(B + A) \\ &= A + B \end{aligned}$$

C, $\bar{C}$ আছে, বাদ
 $A\bar{B}$ থাকবে

$$\bar{C} + A = \bar{C} + A$$

$$F = ABC + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}C$$

3 variable এর জন্য $\rightarrow 2^n$

$$= 2^3$$

$$= 8$$

		BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
		00	01	11	10	
A	0		1	1		
A	1		1	1		

কম্প্লিমেন্ট
Area
(A+B)(B+B)

$$F = C$$

A ও $\bar{A}$ আছে
B ও $\bar{B}$ আছে

কিন্তু C কম্প্লিমেন্ট একই আছে
এ নাই

So, Answer হবে = C

$$F = \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$$

		BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
		00	01	11	10	
A	0				1	
A	1	1	1	1	1	

($B\bar{C}$)
A ও $\bar{A}$ আছে
 $B\bar{C}$ আছে

এখানে কম্প্লিমেন্ট A-ই আছে

$$F = A + B\bar{C}$$

$$F = \bar{A}BC + A\bar{B}\bar{C} + AB\bar{C} + ABC$$

		BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
		00	01	11	10	
A	$\bar{A}$	0				
1	0			1		
1	1	1		1	1	

A ও $\bar{A}$ আছে
BC আছে

$A\bar{B}\bar{C}$

A

A

$AB\bar{C}$

Same

$\bar{C}A + BC$

বুঝাও যা যা আছে

$AB\bar{C}$

01 11 10 00

A আছে

$\bar{C}$ আছে

B ও $\bar{B}$ আছে

AC

$$F = x'y'z' + x'yz' = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C}$$

		BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
		00	01	11	10	
A	$\bar{A}$	0				
1	0	1				
1	1					

B ও $\bar{B}$ আছে, বাদ
 $\bar{A}\bar{C}$ থাকবে শুধু

যা যা আছে তাই বুঝাও

$$F = x'yz + x'y'z' + xyz + xyz'$$

		$x'yz$	$x'y'z'$	yz	yz'
		00	01	11	10
x	x'	0			
1	0			1	
1	1	1		1	1

$x'yz'$ $x'yz$ yz yz'

z, z' আছে, বাদ (x, y)

$x'x'$ বাদ $x'yz' + yz' = F$

$x'yz'$ $x'yz$

$$F = yz + xz'$$

y, y' বাদ

xz'

$$F = ABCD + AB\bar{C}\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + ABC\bar{D}$$

4 variable $\rightarrow 2^4 = 16$ ফলাফল

		CD			
		$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	$C\bar{D}$ 11	CD 10
AB	$\bar{A}\bar{B}$ 00				1
	$\bar{A}B$ 01				1
	AB 11	1	1	1	1
	$A\bar{B}$ 10		1		

এখানে AB আছে
C, $\bar{C}$, D, $\bar{D}$ সবই আছে
তাই C, $\bar{C}$ ও D, $\bar{D}$ বাদ
থাকবে শুধু AB

AB, $A\bar{B}$ ও $\bar{C}D$ আছে
(AB)

B, $\bar{B}$ আছে, তাই এটা বাদ
 $A\bar{C}D$ থাকবে শুধু
($A\bar{C}D$)

$$F = AB + A\bar{C}D$$

Canonical / Standard form থেকে অথবা K-Map দিয়ে Simplification

$$F(A, B, C, D) = \sum m(0, 7, 9, 12, 15)$$

$$2^4 = 16$$

Sum of Minterms/
SOP

High signal

4 bit

8	4	2	1
---	---	---	---

$$0 = 0000 = \bar{A}\bar{B}\bar{C}\bar{D}$$

$$7 = 0111 = \bar{A}BCD$$

$$9 = 1001 = A\bar{B}\bar{C}D$$

$$12 = 1100 = AB\bar{C}\bar{D}$$

$$15 = 1111 = ABCD$$

$$F(A, B, C, D) = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}D + AB\bar{C}\bar{D} + ABCD$$

		CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
		00	01	11	10	
AB						
$\bar{A}\bar{B}$	00	1				
$\bar{A}B$	01			1		
$A\bar{B}$	11	1		1		
AB	10		1			

$\left. \begin{matrix} \bar{A}B \\ AB \\ CD \end{matrix} \right\} A \text{ ও } \bar{A} \text{ বাদ}$
 $B\bar{C}D$

29-08-2023

Convert

*Identify the variables

*Identify the missing variables

*Apply $\rightarrow$ Boolean Algebra

$$\begin{array}{ll} \text{SOP} \rightarrow & A + A' = 1 \\ & B + B' = 1 \\ & C + C' = 1 \end{array} \quad \begin{array}{l} \text{POS} \rightarrow AA' = 0 \\ BB' = 0 \\ CC' = 0 \end{array}$$

Minimal form $\rightarrow$ সব variable থাকবে না

Canonical form $\rightarrow$ সব variable থাকবে

Minimal to canonical form Convert

SOP

$$F = B + \bar{A} \bar{C}$$

$$A + \bar{A} = 1$$

$$B + \bar{B} = 1$$

$$C + \bar{C} = 1$$

$$= B(A + \bar{A}) + \bar{A} \bar{C}(B + \bar{B})$$

$$= AB + \bar{A}B + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$= AB(C + \bar{C}) + \bar{A}B(C + \bar{C}) + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$= ABC + AB\bar{C} + \bar{A}BC + \bar{A}B\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$= ABC + AB\bar{C} + \bar{A}BC + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

POSOP

$$0 = \bar{A} \cdot A$$

$$B + \bar{A} \bar{C} = \bar{B} \cdot B$$

$$(B + \bar{A}) (B + \bar{C}) \text{ (Maxterm)} \cdot (A + \bar{C} + B) \cdot (\bar{C} + B + \bar{A}) =$$

$$= (B(C + \bar{C}) + \bar{A}(C + \bar{C})) \cdot (B(A + \bar{A}) + \bar{C}(A + \bar{A})) \cdot (\bar{C} + B + \bar{A}) =$$

$$= (BC + B\bar{C}) + \bar{A}C + \bar{A}\bar{C} \cdot (AB + \bar{A}B) + \bar{A}\bar{C} + \bar{A}\bar{C} \cdot (\bar{C} + B + \bar{A}) =$$

POS

$$\rightarrow \boxed{A\bar{A}=0 \quad B\bar{B}=0 \quad C\bar{C}=0}$$

$$F = (A + B + C') (A' + C)$$

$$= (A + B + C') (A' + C + 0)$$

$$= (A + B + C') (A' + C + B \cdot B')$$

$$= (A + B + C') \left(\frac{A' + C + B}{A + B} \right) \left(\frac{A' + C + B'}{A + C + B} \right)$$

$$A + BC = (A + B)(A + C)$$

Apply Distributive law

$$= AB(C + \bar{C}) + A\bar{B}(C + \bar{C}) + \bar{A}B(C + \bar{C}) + \bar{A}\bar{B}(C + \bar{C})$$

$$= ABC + AB\bar{C} + \bar{A}BC + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$= ABC + AB\bar{C} + \bar{A}BC + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

POS Minimal to Canonical form

$$(B + \bar{A}) (B + \bar{C})$$

$$\begin{array}{l} A \cdot \bar{A} = 0 \\ B \cdot \bar{B} = 0 \\ C \cdot \bar{C} = 0 \end{array}$$

$$= (\bar{A} + B + \underline{C \cdot \bar{C}}) (B + \bar{C} + A \cdot \bar{A})$$

$$= (\bar{A} + B + \underline{C}) (\bar{A} + B + \bar{C}) (B + \bar{C} + A) (B + \bar{C} + \bar{A}) \rightarrow \text{Apply distributive Law}$$

$$= (\bar{A} + B + C) (\bar{A} + B + \bar{C}) (A + B + \bar{C}) (\bar{A} + B + \bar{C})$$

$$= (\bar{A} + B + C) (\bar{A} + B + \bar{C}) (A + B + \bar{C})$$

SOP

$$= A + \bar{B}C$$

$$= A(B + \bar{B}) + \bar{B}C(A + \bar{A})$$

$$= AB + A\bar{B} + A\bar{B}C + \bar{A}BC$$

$$= AB(C + \bar{C}) + A\bar{B}(C + \bar{C}) + A\bar{B}C + \bar{A}BC$$

$$= ABC + AB\bar{C} + \underline{A\bar{B}C} + A\bar{B}\bar{C} + \underline{A\bar{B}C} + \bar{A}BC$$

$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}BC$$

POS
 $(A + \bar{B}) (A + C)$

$$\begin{aligned} & (A + \bar{B} + C\bar{C}) (A + B\bar{B} + C) \\ &= (A + \bar{B} + C) (A + \bar{B} + \bar{C}) (A + B + C) (A + \bar{B} + C) \\ &= (A + \bar{B} + C) (A + \bar{B} + \bar{C}) (A + B + C) \end{aligned}$$

DLT 2003-00-20

POS
 $(A + B + \bar{C}) (\bar{A} + C)$

$$\begin{aligned} &= (A + B + \bar{C}) (\bar{A} + B\bar{B} + C) \\ &= (A + B + \bar{C}) (\bar{A} + B + C) (\bar{A} + \bar{B} + C) \end{aligned}$$

Redundancy Theorem

* $AB + \bar{A}C + BC$ # 3 variable এর

A	B	C
0	1	0

$AB + \bar{A}C$ complement থাকলে এবং একই না থাকলে

যে variable এর complement নাই সেই চ্যক

অবস্থিত term দুটে নিব। অন্যটি বাদ যাবে।

* $\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C} \rightarrow$ বাদ
 $= \bar{A}\bar{B} + \bar{A}\bar{C}$ A, A'

* $\bar{A}B + \bar{A}\bar{C} + \bar{B}\bar{C}$
 $= \bar{A}B + \bar{B}\bar{C}$

* $\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}C$
 $= \bar{A}\bar{C} + \bar{B}C$

$A + \bar{A}B$
 $= (A + \bar{A})(A + B)$
 $= (A + B)$

$$A + \bar{A}B$$

$$(A + \bar{A}) + (A \cdot B)$$

$$= A\bar{A} + AB + \bar{A}B + AA$$

$$= (0 + A + AB + \bar{A}B) = A + B(A + \bar{A})$$

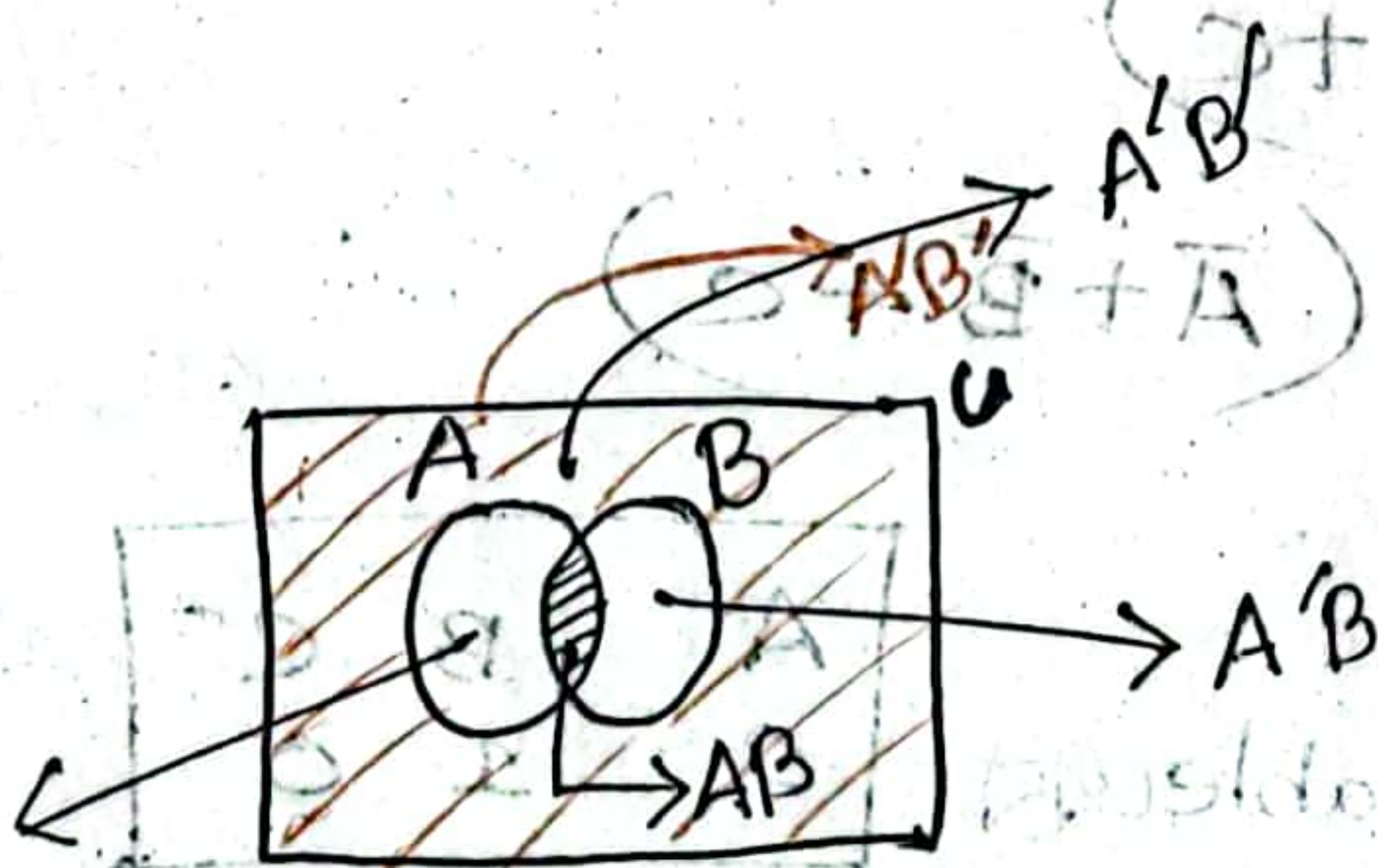
$$= A(1 + B) + \bar{A}B = A + B$$

02-09-2023

DLD

statement

Venn Diagram



$$A \cap B \rightarrow (A \cdot B) \quad (U \setminus A) \rightarrow \bar{A} \quad (U \setminus B) \rightarrow \bar{B}$$

$$A \setminus B \rightarrow A \cdot \bar{B}$$

$$B \setminus A \rightarrow B \cdot \bar{A}$$

$$A \cup B \rightarrow (A + B)$$

$\cap \rightarrow$ AND

$\cup \rightarrow$ OR

$\bar{} \rightarrow$ complement

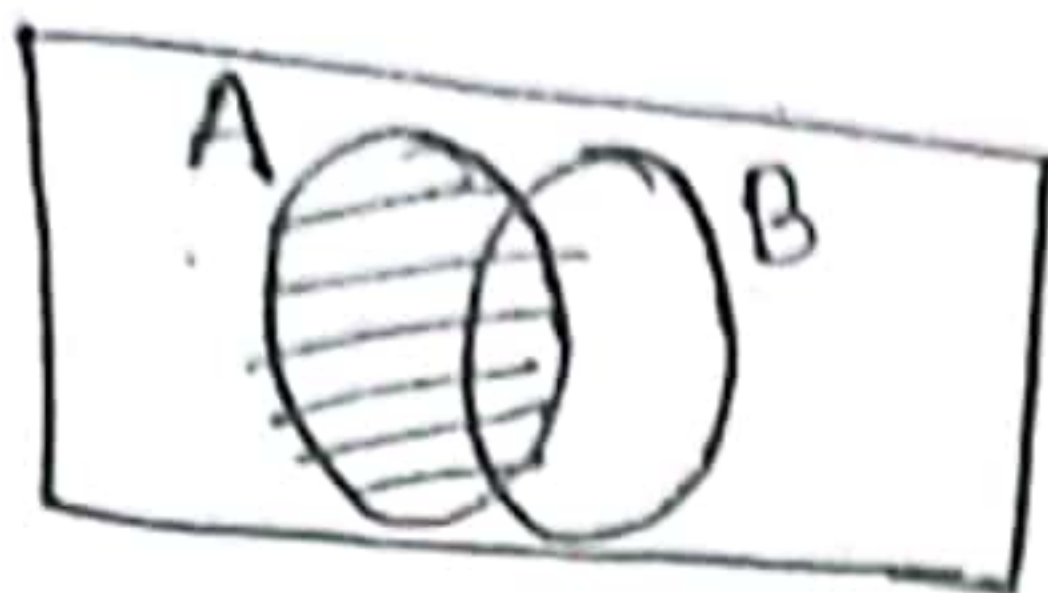
$$A + \bar{A}B$$

$$(A + B)(\bar{A} + A)$$

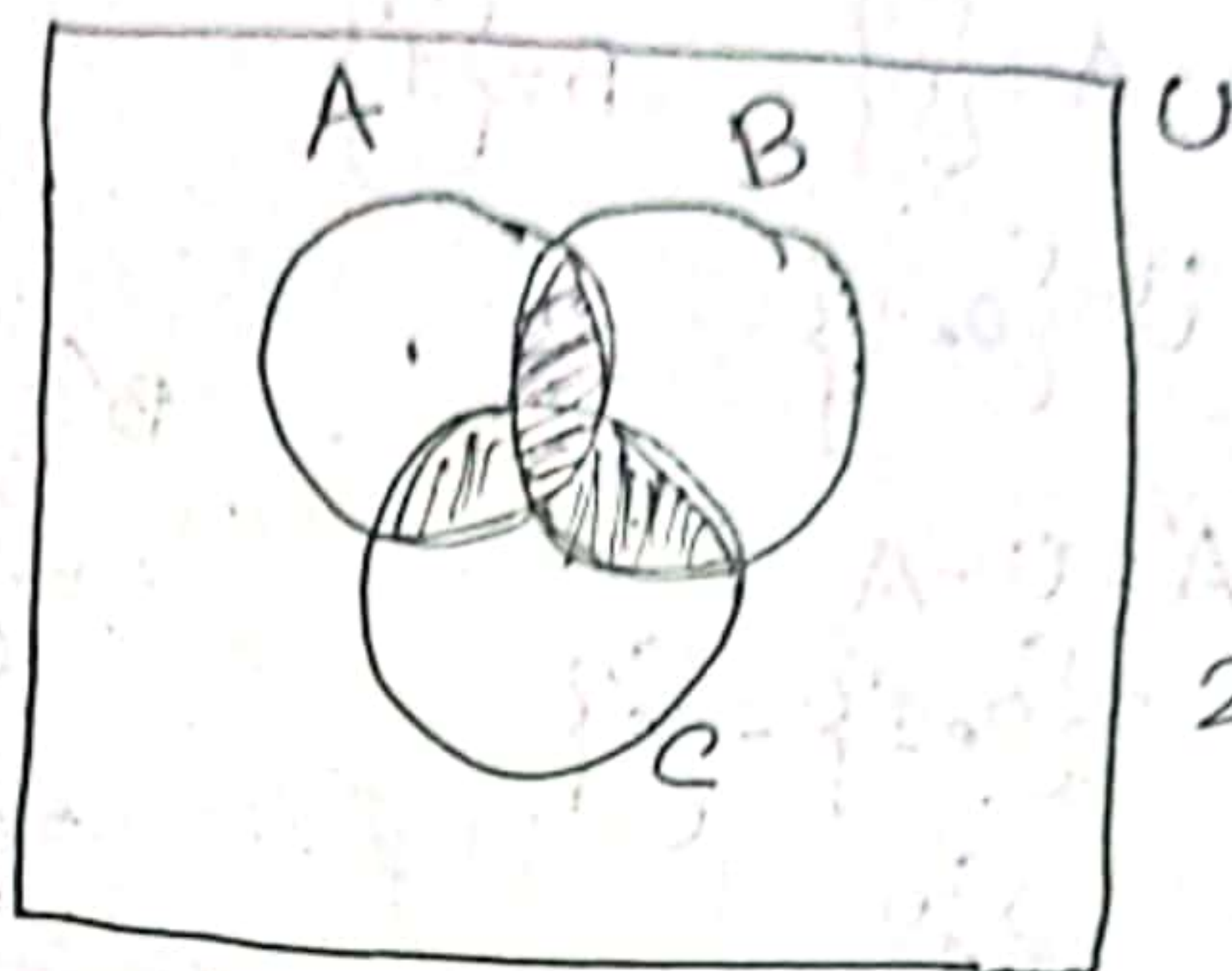
$$(A + B) =$$

$$A\bar{B} + \bar{A}B + \bar{A}\bar{B}$$

$$= A\bar{B} + B\bar{A} + \bar{A}\bar{B}$$



$$\begin{aligned} & A\bar{B} + AB \\ &= A(\bar{B} + B) \\ &= A \end{aligned}$$



$$2^3 = 8$$

$$\begin{aligned} & AB\bar{C} + A\bar{B}C + \bar{A}BC + ABC \\ &= AB(\bar{C} + C) + A\bar{B}C + \bar{A}BC = \bar{A}\bar{B}C \\ &= AB + A\bar{B}C + \bar{A}BC \\ &= AB + C(A\bar{B} + \bar{A}B) \quad \text{Statement} \\ &= AB + C(A \oplus B) \end{aligned}$$

* F will be high if the following statement is high.

i) B and C $\xrightarrow{T} BC$

ii) A and C $\xrightarrow{F} \bar{A}\bar{C}$

iii) A, B, C $\xrightarrow{T} ABC$

iv) $((A \text{ and } C) T) F \rightarrow \overline{A \cdot C}$

$$= BC + A' + C'$$

$$= (B + C')(C + C') + A' = A' + B + C'$$

$$\begin{aligned} & BC + A'C' + ABC + \bar{A}\bar{C} \\ &= BC + A'C' + ABC + A' + C' \\ &= BC + A'(C' + 1) + ABC + C' \\ &= BC(1 + A) + A' + C' \end{aligned}$$

A or C $\xrightarrow{F} \bar{A} + \bar{C}$

$\{0,1\}$ Binary Univers

$A = \{0\}$ $B = \{1\}$

$U = \{0,1\}$

$A' = U - A$
 $= \{0,1\} - \{0\}$
 $= \{1\}$

$B' = U - B$

$= \{0,1\} - \{1\}$
 $= \{0\}$

r 's Complement

যে কোনো সংখ্যা পদ্ধতির Base (ভিত্তি)

Binary $\rightarrow 2$'s $\xrightarrow{1 \text{ বাদ দিলে}}$ 1's

Decimal $\rightarrow 10$'s $\xrightarrow{1 \text{ বাদ দিলে}}$ 9's $\rightarrow 10$ এর complement

থেকে 1 বাদ দিলে 9 এর complement আসবে

সূত্র:

$(r^n - N)$

n = Number of Digit

দশমিকের আগে কতগুলো

Digit আছে যেটা



Integer Part-এ

r = Base

n = Number of digit

N = যে Number এর complement বের করা যাবে

$(5)_{10} = r^n - N = (10)^1_{10} - (5)_{10}$
 $= 10 - 5$
 $= (5)_{10}$

$$(1011)_2$$

$$r^n - N$$

$$= (2^4)_{10} - (1011)_2$$

$$= (16)_{10} - (1011)_2$$

$$= (10000)_2 - (1011)_2$$

$$= (0101)_2$$

m = দক্ষিণের পর কয়টি Digit আছে

$$r'S = r^n - N$$

$$(r-1)'S = r^n - N - 1 \quad / \quad r^n - r^m - N$$

$$= (2^4)_{10} - (1011)_2 - (1)_{10} = (16)_{10} - (1011)_2 - (1)_{10}$$

$$= (10000)_2 - (1011)_2 - (1)_{10} = (16-1)_{10} - (1011)_2$$

$$= (1111)_2 - (1011)_2 = (15)_{10} - (1011)_2$$

$$= (0100)_2$$

$$(5670)_{10}$$

$$= (10^4)_{10} - (5670)_{10}$$

$$= (4330)_{10}$$

$$(10^4)_{10} - (5670)_{10} - (1)_{10}$$

$$= (9999)_{10} - (5670)_{10}$$

$$= (4329)_{10} \text{ } (r-1)'s \text{ complement}$$

$$\begin{array}{r} 4329 \\ +1 \\ \hline 4330 \end{array} \rightarrow r's \text{ complement}$$

$$*(r-1)'s \text{ complement} + 1 = r's \text{ complement}$$

Data Representation

Magnitude

Complement

Signed

Unsigned

1's

2's

+ve, -ve

+ve, -ve

Binary Number

(+, -)ve

(+ve) Binary number

Binary number

Unsigned

Unsigned

only positive

$$+9 = 01001$$

$$-9 = 11001$$

Sign bit

MSB

0 থেকে $(2^n - 1)$

পর্যন্ত সংখ্যা

$$+6 = 110$$

-6 = Can't representation

Signed (+, -) both

$$+6 = 0110$$

$$-6 = 1110$$

Sign bit

magnitude

Sign bit

Sign bit

0 = positive number

1 = Negative number

$$\text{Range} \Rightarrow -2^{n-1} \rightarrow 2^{n-1} - 1$$

if, $n=4$

(-7 থেকে +7)

$$2^{4-1} - 1$$

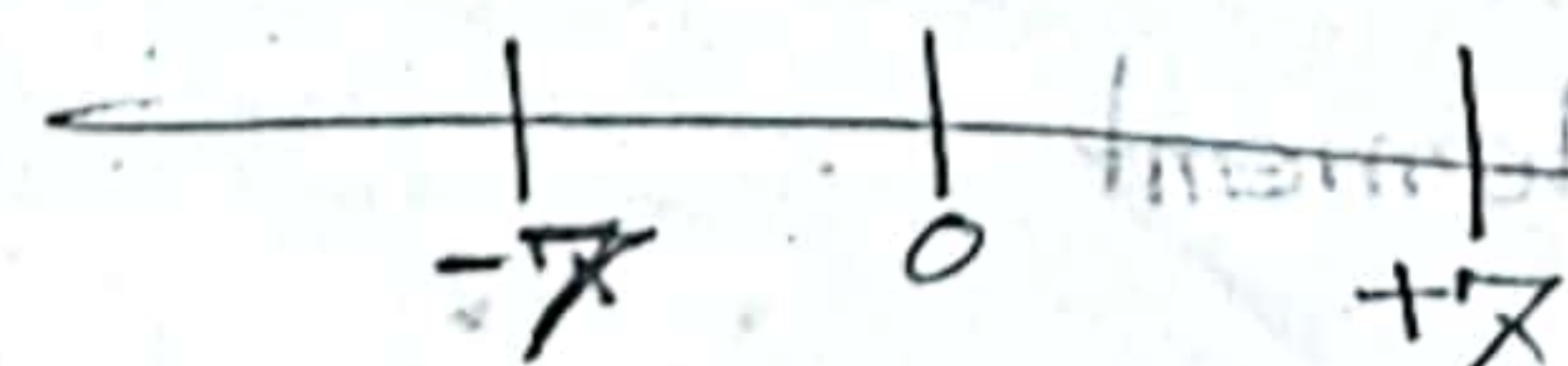
$$= 2^3 - 1$$

$$= 8 - 1$$

$$= 7$$

$$-7 \rightarrow +7$$

এখানে Total 15 টি সংখ্যা



0 (Zero) থেকে 2 টি একটা +0

অপরটি -0

Use Unsigned magnitude

+0 → 0 0000
 -0 → 1 0000
 ↘ sign bit

1) +5 and -5
 +5 = 0101
 -5 = 1 0101
 ↘ sign bit

১) +9 and -9

+9 = 01001
 -9 = 1 01001
 ↘ sign bit

Unsigned → 1 1 1 0 1 (only +ve)
 (-13) = 1 1 1 0 1 → Not valid value
 ↘ sign bit

III) +16 and -16
 +16 = 010000
 -16 = 1 010000
 ↘ sign bit

1's complement

1 1 1 0 1
 (-) 0 0 0 1 0 → 1's complement

1's complement করার পর যদি 1 অংক্যা বেড়ে যায়/

আমনে 1 আছে — অংক্যাটি positive
 এবং 1 যোগ করতে হবে

আমনে 1 বাদে বাকি
 যে Digit গুলো
 থাকবে সেগুলোর সাথে

last এ যে 1 আছে সেই 1 কে
 result এর সাথে যোগ করব।

* 0 আছে না আমনে/ — অংক্যাটি negative
 অংক্যাটিকে আবার 1's

আমনে 0 আমনে
 16 8 4 2 1

complement করতে হবে

i) +10 = 1010
 (-10) 0101 — 1's

এক ওটাই result.

III) +23 = 00010111

(-23) = 11101000 — 1's

II) +4 = 0100
 (-4) = 1011 — 1's

2's complement

$$+6 = 0110$$

$$1001 \text{ --- } 1's$$

$$+1$$

$$(-6) \rightarrow 1010 \text{ --- } 2's$$

$$(-2^{n-1}) \text{ to } (2^{n-1}-1)$$

সংখ্যা 1 আমলে সংখ্যা $\rightarrow$ Positive এবং সংখ্যা বিবর্তন নয়।

সংখ্যা(0) আমলে/1না আমলে $\rightarrow$ Negative সংখ্যা।

sign bit $\rightarrow 1 \rightarrow$ Negative
 $0 \rightarrow$ positive

সংখ্যাকে আবার 2's complement করতে হবে।

r's Complement / Radix complement

Given a positive Number N in base r with an integer part of n digits, the r 's complement of

N is defined as $r^n - N$ for $N \neq 0$ and

0 for $N = 0$ [N এর মান 0 আমলে রেজাল্ট 0]

$r =$ Base (ভিত্তি)

ডেসিমালের ক্ষেত্রে r 's complement
 মানে 10's complement

$N =$ যে Number-টা দেওয়া থাকবে যেটা

Bimary-র ক্ষেত্রে r 's complement

$n =$ Number of digit

মানে 2's complement

$$(1201)_{10}$$

$\rightarrow$ এখানে চারটি

Digit আছে।

ইন্টিজার পাঁচে

কতটি Digit আছে

e.g. → The 10's complement of $(52520)_{10}$

$$\Rightarrow (52520)_{10}$$

Base $r=10$

Digit $n=5$

$$(52520)_{10} = (10)^5 - (52520)_{10}$$

$$(001101)_{10} - (000000)_{10} = (001101)_{10}$$

$$N = 52520$$

$$\Rightarrow (0.3267)_{10}$$

Base $r=10$

Digit $n=0$

$$N = 0.3267$$

→ দশমিকের আগে কোনো integer part
নাই। অর্থাৎ $n=0$

$$r^n - N$$

$$(0.3267)_{10} = (10)^0 - (0.3267)_{10}$$

$$(01)_{10} - (0.3267)_{10}$$

$$= (0.6733)_{10}$$

e.g. → 10's complement of $(25.639)_{10}$

$$\Rightarrow (10)^2 - (25.639)_{10}$$

$$= (100)_{10} - (25.639)_{10}$$

$$= (74.361)_{10}$$

$$r=10$$

n = দশমিকের আগে Integer
part এ 2টি digit আছে

$$\text{অর্থাৎ } \boxed{n=2}$$

eg $\rightarrow$ 2's complement of $(101100)_2$

$$\Rightarrow \boxed{r^n - N}$$

$$(101100)_2 = (2^6 - (101100)_2)$$

$$= (64)_{10} - (101100)_2$$

$$= (1000000)_2 - (101100)_2$$

$$= (10100)_2$$

$$\begin{array}{r} 1000000 \\ - 101100 \\ \hline 0010100 \end{array}$$

eg $\rightarrow$ 2's complement of $(00110)_2$

$$\Rightarrow \text{2's complement of } (0.0110)_2 = r^n - N = (1.0000)_2 - (0.0110)_2$$

$$r=2$$

$$n=0$$

$$N=0.0110$$

$$\begin{array}{r} 1.0000 \\ - 0.0110 \\ \hline 0.1010 \end{array}$$

$$= (2^0)_{10} - (0.0110)_2$$

$$= (1)_{10} - (0.0110)_2$$

$$= (1.0000)_2 - (0.0110)_2$$

$$= (0.1010)_2$$

$$\boxed{S=0}$$

$(r-1)$'s Complement $r^n - r^{-m} - N$

Given a positive number N in base r with an integer part of n digits and a fraction part of m digits, the $(r-1)$'s complement of N is defined as $(r^n - r^{-m} - N)$

$r = \text{base}$

$n = \text{Number of Digit (an integer part)}$

$m = \text{দশমিকের পর কয়টি Digit আছে}$

$(r-1)$'s complement $\rightarrow$ ডেসিমালের ক্ষেত্রে 9's complement হবে। $\Rightarrow (10-1)=9$

$\rightarrow$ বাইনারির ক্ষেত্রে 1's complement হবে। $\Rightarrow (2-1)=1$

e.g

* The 9's complement of $(52520)_{10} = ?$

$\Rightarrow$ The 9's complement of $(N)_{10} = r^n - r^{-m} - N$

$(52520)_{10}$ এর 9's complement $= 10^5 - 10^0 - 52520$

$n=5$

$m=0$

$$\begin{aligned} &= (100000)_{10} - (1)_{10} - (52520)_{10} \\ &= (47479)_{10} \end{aligned}$$

{ কোনো কিছুর power $\rightarrow 0$ হলে
মান হয় 1

eg $\rightarrow$ The 9's complement of $(0.3267)_{10}$

$$\begin{aligned}\Rightarrow (0.3267)_{10} &= \left(\frac{10^0}{10}\right) - \left(\frac{10^{-4}}{10}\right) - (0.3267)_{10} \\ &= (1)_{10} - (0.0001)_{10} - (0.3267)_{10} \\ &= (0.9999)_{10} - (0.3267)_{10} \\ &= (0.6732)_{10}\end{aligned}$$

eg $\rightarrow$ 1's complement of $(101100)_2 = ?$ $(r^m - r^m - N)$

$$\begin{aligned}\Rightarrow (101100)_2 &= (2^6)_{10} - (2^0)_{10} - (101100)_2 \\ &= (64)_{10} - (1)_{10} - (101100)_2 \\ &= (1000000)_2 - (1)_{10} - (101100)_2 \\ &= (64-1)_{10} - 101100 \\ &= (111111)_2 - (101100)_2\end{aligned}$$

$$\begin{array}{r} 111111 \\ - 101100 \\ \hline 010011 \end{array} \Rightarrow (10011)_2$$

eg $\rightarrow$ 1's complement of $(0.0110)_2$

$$\begin{aligned}\Rightarrow (0.0110)_2 &= (2^0)_{10} - (2^{-4})_{10} - (0.0110)_2 \\ &= (1)_{10} - (0.0625)_{10} - (0.0110)_2 \\ &= (1-0.0625)_{10} - (0.0110)_2 \\ &= (0.9375)_{10} - (0.0110)_2\end{aligned}$$

03-09-2023

2's Complement

15 - 22

1111

10110

$$15 + (-22) = (-7)$$

01001

1's complement

01111
00000
00000

01111
01001

11000

00111

10001

(-7)

8 Bit নিয়ে কাজ করতে হবে.

16 8 4 2 1
1 1 1 1

15 - 22

10110

15 - 1111

10110

+22 01001

+1

01010 2's

1111

1010

00001

sign bit
Carry bit নাই

+22 → 00010110

11101001 1's

+1

(-22) 11101010 2's

15 → 00001111

(+) -22 → 11101010

(-7) 11111001

Result

bit সংখ্যা বাড়েনি

মান নেও আছে (করেনি)

Result এর 2's complement

Result এর 1's complement

করে 1 লাগ করতে হবে

00000110

+1

00000111

(Sign bit)

অংখ্যাটি Negative

Using 2's complement

$$15 - 22 = ?$$

Binary value

$$15 = 1111$$

$$22 = 00010110$$

$$11101001 - 1's$$

+1

$$(-22) = 11101010 - 2's$$

$$\begin{array}{r} 1111 \\ 1010 \\ \hline 11001 \end{array}$$

আমনে 0
আছে

(কোনো extra carry নেই)

$$15 \rightarrow 00001111$$

$$(+)-22 \rightarrow 11101010 - 2's$$

$$11111001 \rightarrow \text{result}$$

$$00000110 \rightarrow \text{result এর 1's complement}$$

+1

$$00000111$$

এর সাথে 1 যোগ

Sign bit 0 (-ve)

বসতে হবে

BCD — 4 bit — এর কোড

Binary Coded Decimal

BCD $\rightarrow$ D

D $\rightarrow$ BCD

$$15 + 22$$

$$= 37$$

BCD

$$\begin{array}{r} 15 + 27 \\ \swarrow \quad \downarrow \quad \searrow \\ 0001 \quad 0101 \quad 0010 \quad 0111 \end{array}$$

$$5 + 2$$

$$1 + 2$$

$$\begin{array}{r} (15+22) \text{ BCD} \rightarrow \text{(BCD কোড)} \\ 0001 \ 0101 \\ 0010 \ 0111 \\ \hline 0011 \ 1100 \end{array}$$

$$\begin{array}{r} 3 \quad 12 \end{array}$$

$$\rightarrow 1279$$

and final carry = 0

Binary কোড

$$00001111 - 15$$

$$00100101 - 37$$

$$00110100 - 52$$

Binary Sum

$$0000 \ 1111$$

$$0010 \ 1010$$

$$0101$$

$$0010$$

$$0111$$

$$15 + 37$$

$$= 42$$

$$64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1$$

$$37 \rightarrow 1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \text{ } \left. \begin{array}{l} \text{bin} \\ \text{val} \end{array} \right\}$$

$$42 \rightarrow 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \text{ } \left. \begin{array}{l} \text{bin} \\ \text{val} \end{array} \right\}$$

$$22 \rightarrow 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \text{ } \left. \begin{array}{l} \text{bin} \\ \text{val} \end{array} \right\}$$

$$27 \rightarrow 0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0 \text{ } \left. \begin{array}{l} \text{bin} \\ \text{val} \end{array} \right\}$$

$$5 \text{ এর BCD value} = 0101$$

$$2 \text{ ————— } 0010$$

$$1 \text{ ————— } 0001$$

$$2 \text{ ————— } 0010$$

$$7 \text{ ————— } 0111$$

3 condition BCD

- 9 থেকে ছোট এবং carry 1 পালে $\rightarrow$ 6 যোগ করা (0110)
 9 থেকে ছোট এবং carry 0 পালে $\rightarrow$ Correct
 9 > — carry না থাকলে 6 যোগ করা (0110)

BCD যোগ

$(15)_{10} + (37)_{10} = (52)_{10}$
 1 3 5 7
 0001 0011 0101 0111

15 000 10101
 37 001 10111
 —————
 010 01100
 4 12

$12 > 9$
 Add extra (0110)
 01001100
 0110
 —————
 01010010
 5 2

Gray Code

$(15)_{10} + (22)_{10} = (37)_{10}$

15 — 0001 0101
 22 — 0010 0010
 —————
 0011 0111
 3 7

$(37)_{10}$ বা, $(00110111)_{BCD}$

1 $\rightarrow$ 0001 3 $\rightarrow$ 0011
 5 $\rightarrow$ 0101 7 $\rightarrow$ 0111

01001100
 00000110 Add extra (0110)
 —————
 01010010

$(01010010)_{BCD}$
 $(52)_{10}$ (Ans)

$(15)_{10} + (27)_{10} = (42)_{10}$

15 — 0001 0101
 27 — 0010 0111

0011 1100
 3 12 > 9 and final
 carry = 0

So, Add extra (0110)_{BCD}

0011 1100
 + 0110
 —————
 0100 0010
 4 2

$(01000010)_{BCD}$
 $(42)_{10}$ (Ans)

8 4 2 1
Decimal Binary

0 — 0 0 0 0
 1 — 0 0 0 1
 2 — 0 0 1 0
 3 — 0 0 1 1
 4 — 0 1 0 0
 5 — 0 1 0 1
 6 — 0 1 1 0
 7 — 0 1 1 1

8 — 1 0 0 0
 9 — 1 0 0 1
 10 — 1 0 1 0
 11 — 1 0 1 1
 12 — 1 1 0 0
 13 — 1 1 0 1
 14 — 1 1 1 0
 15 — 1 1 1 1

Gray Code

Gray Code

0 0 0 0
 1 0 0 0
 3 0 0 1
 2 0 0 1
 6 0 1 1
 7 0 1 1
 5 0 1 0
 4 0 1 0

10 1 1 0
 11 1 1 0
 15 1 1 1
 14 1 1 1
 12 1 1 0
 13 1 1 0

পাশাপাশি দুইটা Binary
 digit এর মধ্যে যুগ্ম
 মাত্র একটা বিটের
 পার্থক্য থাকবে।

Binary to Gray Code

(1010)₂ → 1
 ↓
 MSB

EX-OR

B → 1 0 1 0
 ↓ ↓ ↓ ↓
 B → 1 1 1 1
 G → 1 0 1 0
 B → 1 1 0 0

B to G code G to B code

B — b_3 b_2 b_1 b_0

$$B \rightarrow G \rightarrow G_3 = b_3$$

$$G_2 = b_2 \oplus b_3$$

$$G_1 = b_1 \oplus b_2$$

$$G_0 = b_0 \oplus b_1$$

B — b_3 b_2 b_1 b_0

G → q_3 q_2 q_1 q_0

Binary to Gray

Gray to Binary

$$b_3 = G_3$$

$$b_2 = G_2 \oplus b_3$$

$$b_1 = G_1 \oplus b_2$$

$$b_0 = G_0 \oplus b_1$$

$$q_3 = b_3$$

$$q_2 = b_3 \oplus b_2$$

$$q_1 = b_2 \oplus b_1$$

$$q_0 = b_1 \oplus b_0$$

B → G

1101
↓ ↓ ↓ ↓
1011

1000
↓ ↓ ↓ ↓
1100

q_3 q_2 q_1 q_0
1001
↓ ↑ ↑ ↑
1110
 b_3 b_2 b_1 b_0

q_3 q_2 q_1 q_0
1010
↓ ↑ ↑ ↑
1100
 b_3 b_2 b_1 b_0

Binary to Gray
B to G

Gray to Binary
G to B

Gray code $\rightarrow$ 1 1 1 1

binary code $\rightarrow$ 1 0 1 0

* Excess-3 code

* Hamming code

Excess-3

BCD এর সাথে (0011) $\rightarrow$ (3) যোগ করলে
যা হবে মোট Excess-3 Code

BCD $\rightarrow$ 4

0100 $\rightarrow$ 4 এর BCD কোড

0011 $\rightarrow$ তিন (extra)

0111 $\rightarrow$ Excess-3

BCD $\rightarrow$ 3

0011 $\rightarrow$ 3 এর BCD

0011 $\rightarrow$ extra (0011)

0110 $\rightarrow$ Excess-3

Final কোড থাকলে $\rightarrow$ আরও 3 যোগ করব
(0011)

Final কোড না থাকলে $\rightarrow$ (0011) বিয়োগ করব

Binary Code

BCD
8 4 2 1
2 4 2 1

→ Weighted Code

value Depend করে position এর উপর

Excess-3

Gray

ASCII

Unicode

0101
2³2²2¹2⁰

2 4 2 1 → Reflective Code বান

এর

10 10 → এই দুটো 1 জন
↓ ↓
2* 2

Decimal

2 4 2 1 Code

Higher order
Lower order

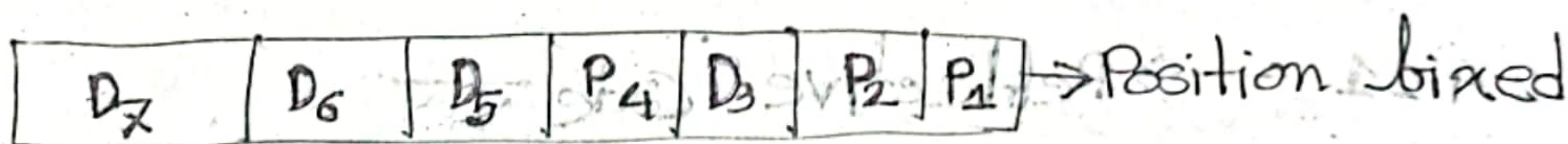
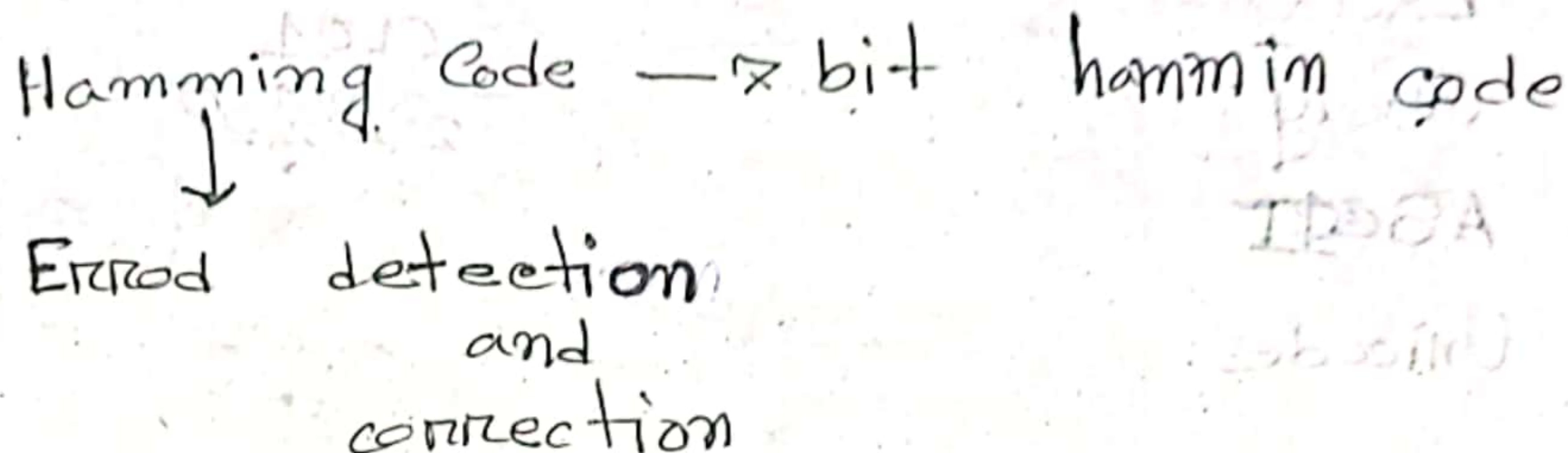
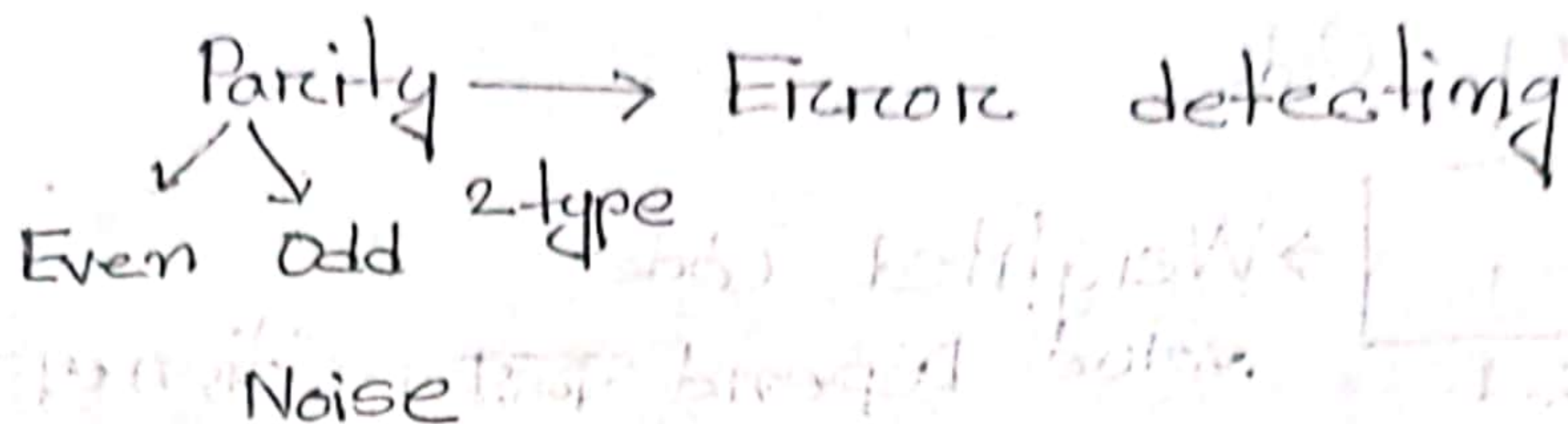
→ 0	0000
→ 1	0001
→ 2	0010 / 1000
→ 3	0011
→ 4	0100 → 1010*
→ 5	1011
→ 6	1100
→ 7	1101
→ 8	1110
→ 9	1111

<u>Decimal</u>	<u>2 4 2 1 code</u>
0	0000
1	0001
2	0010 / 1000
3	0011
4	0100 / 1010
5	1011
6	1100
7	1101
8	1110
9	1111

(0 ও 9 হচ্ছে complement) 1

0 এর complement হচ্ছে 9

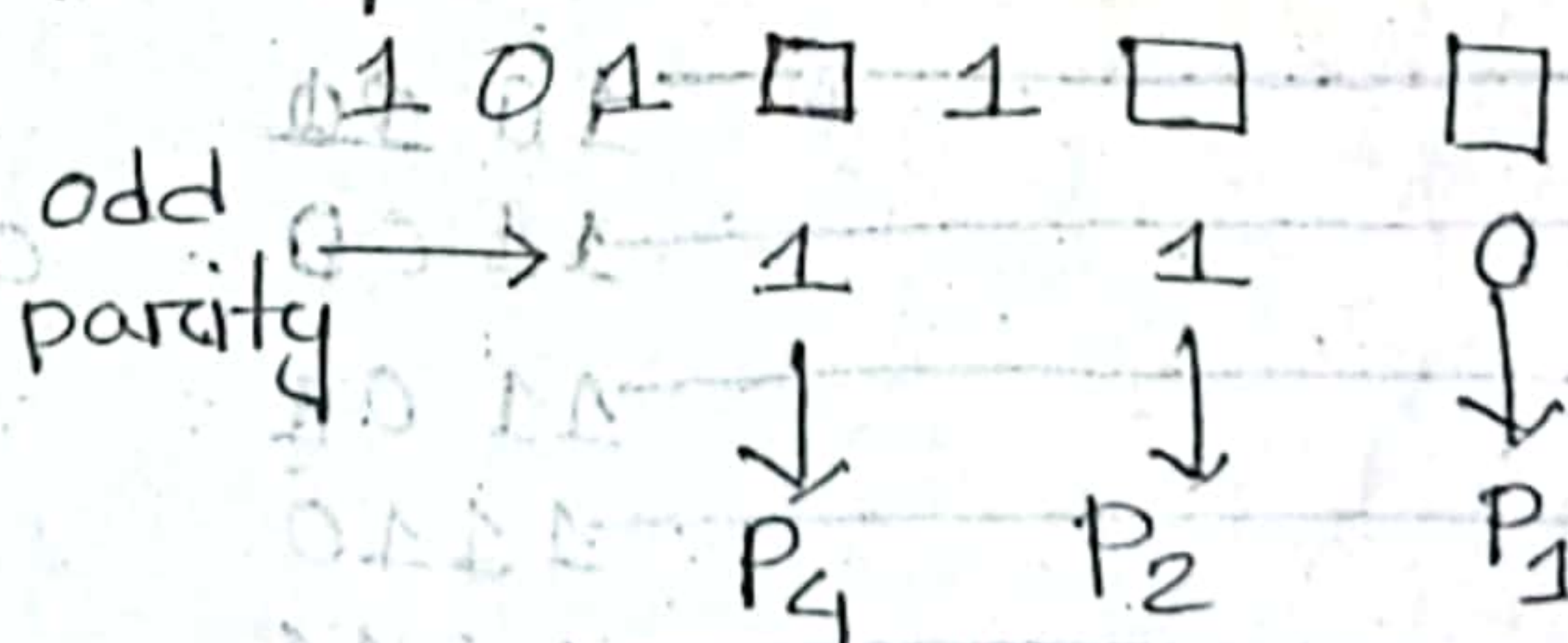
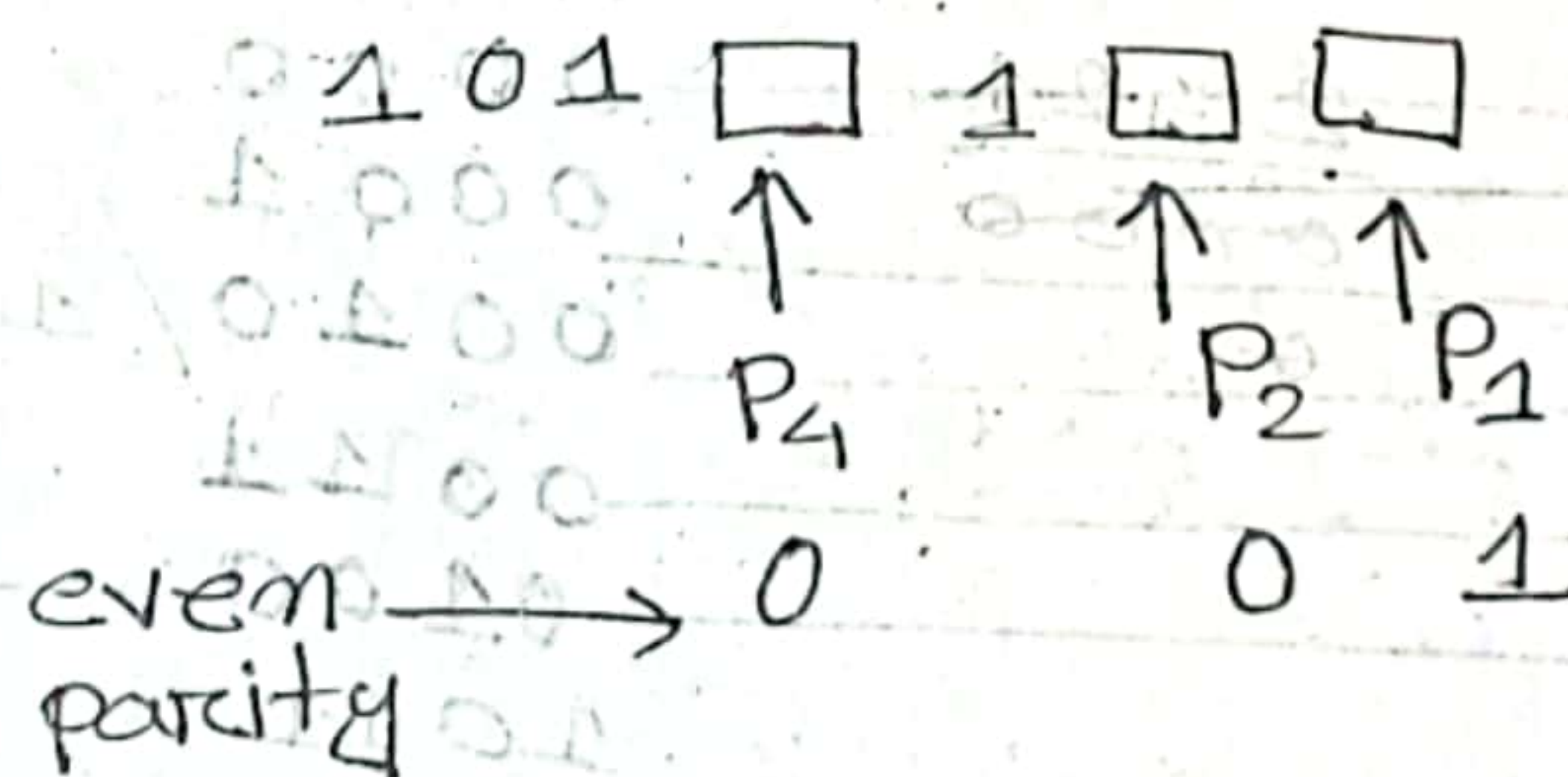
1	8
2	7
3	6
4	5



$$P_1 = D_3 D_5 D_7$$

$$P_2 = D_3 D_6 D_7$$

$$P_3 = D_5 D_6 D_7$$



K Map

09-09-2023

$$* F = x'y'z' + x'y'z + x'y'z' + x'yz'$$

$z \backslash xy$	$x'y'$ 00	$x'y$ 01	xy 11	$x'y'$ 10
0 z'	1	1	1	1
1 z				

x, x' বাদ
 y, y' বাদ

শুধু z' থাকবে

$x \backslash yz$	$y'z'$ 00	$y'z$ 01	yz 11	yz' 10
$x'0$	1			1
$x1$	1			1

x, x' বাদ

y, y' বাদ

z, z' বাদ

A, B, C

$$* F = A'C + A'B + AB'C + BC \rightarrow (SOP) \text{ form}$$

$$= A'C(B+B') + A'B(C+C') + AB'C + BC(A+A')$$

$$= \underline{A'BC} + \underline{A'B'C} + \underline{A'BC} + \underline{A'BC'} + \underline{AB'C} + \underline{ABC} + \underline{A'BC}$$

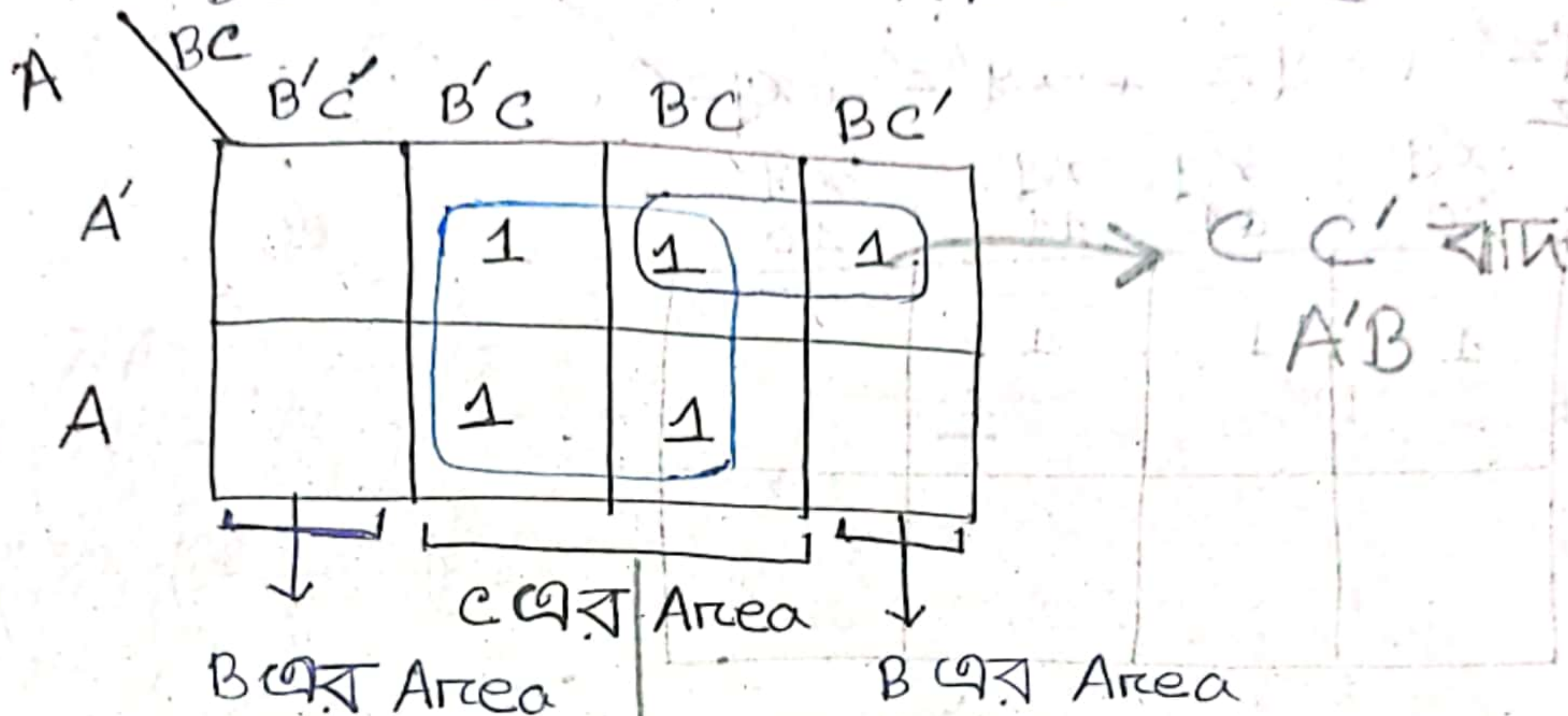
$$= A'B'C + A'BC' + AB'C + ABC + A'BC$$

$$A+A=1$$

$$B+B'=1$$

$$C+C'=1$$

$$F = A'B'C + A'BC' + AB'C + ABC + A'BC$$



$A'A'$ বাদ, $B'B'$ বাদ
C থাকবে শুধু

$$F = A'B + C \quad (\text{Ans})$$

SOP $\rightarrow$ High value নিয়ে কাজ করে
 $F(x, y, z) = \sum (0, 2, 4, 5, 6)$

$$= x'y'z' + x'y'z + x'y'z' + x'y'z + x'y'z'$$

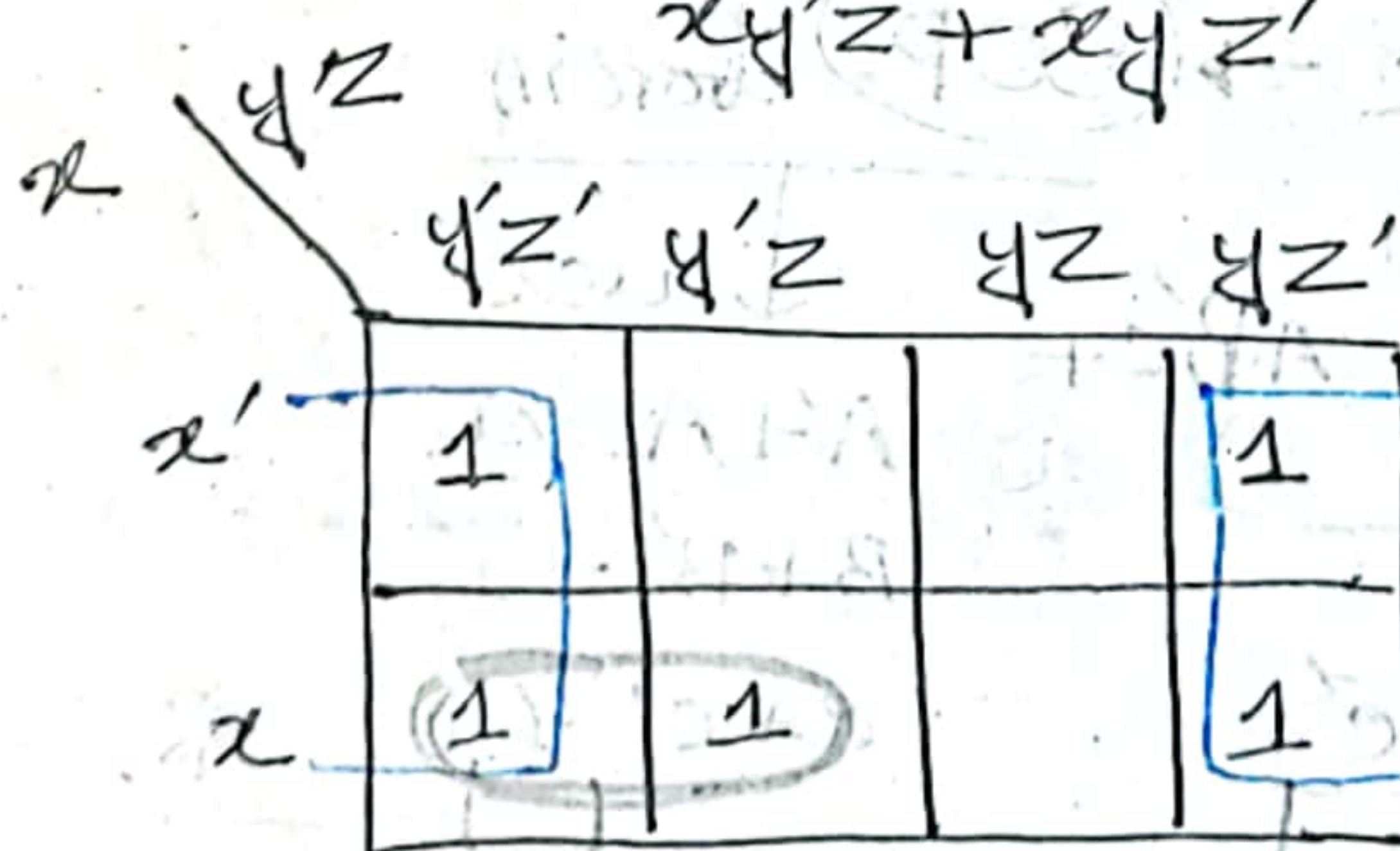
$$000 \rightarrow 0$$

$$010 \rightarrow 2$$

$$100 \rightarrow 4$$

$$101 \rightarrow 5$$

$$110 \rightarrow 6$$



$$F = z' + xy$$

$z'z'$ বাদ
 xy'
 z'

SOP form $\rightarrow$ High value নিচো কাজ কাবে

$$F(w, x, y, z) = \sum (0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

AB \ CD	yz			
	00	01	11	10
w'x'00	1	1		1
w'x01	1	1		1
wx11	1	1		1
wx'10	1	1		1

8 4 2 1

0	0	0	0
1	0	0	1
2	0	0	1
4	0	1	0
5	0	1	0
6	0	1	1
8	1	0	0
9	1	0	0
12	1	1	0
13	1	1	0
14	1	1	1

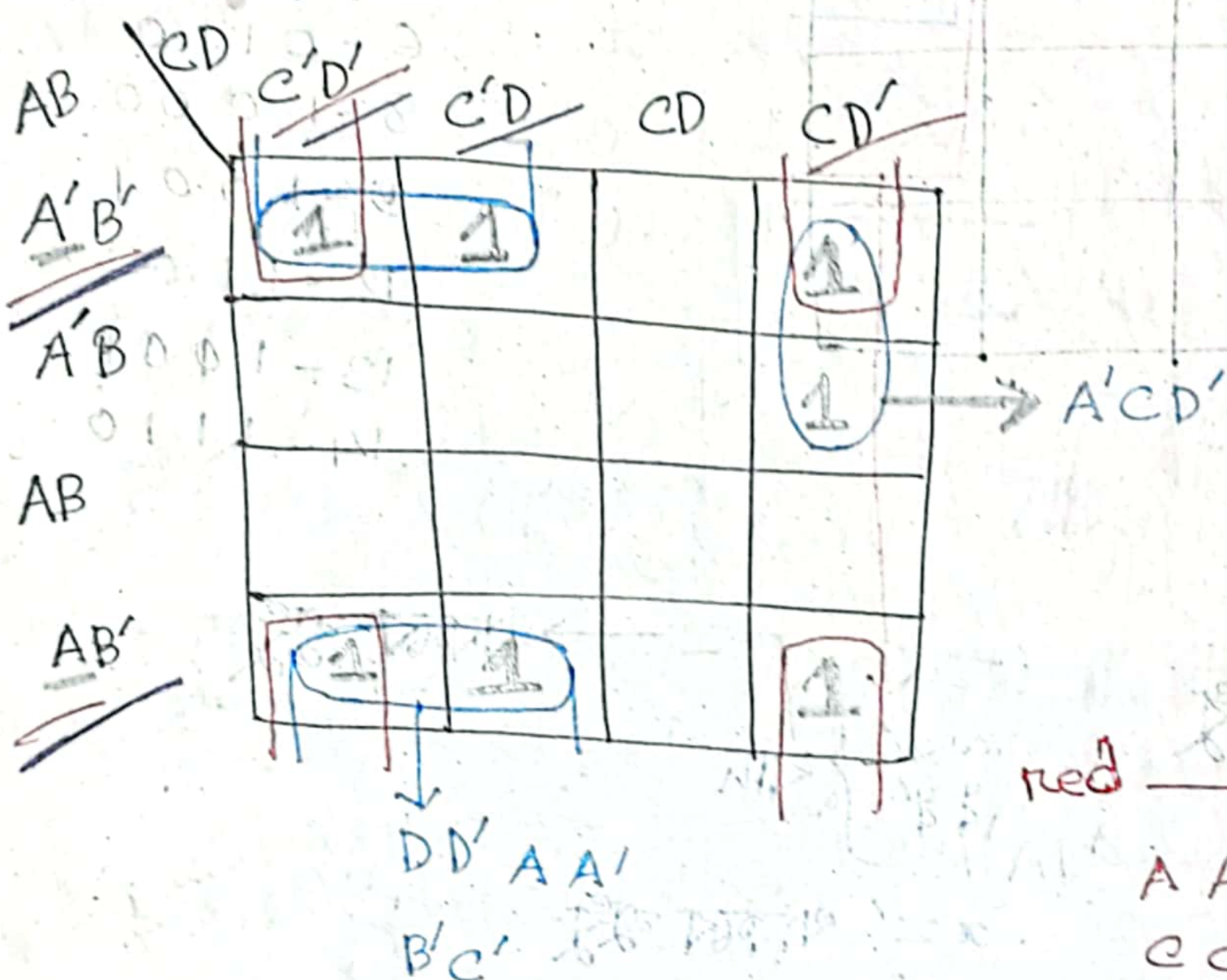
$w'x'00$ বাদ
 $w'x01$ বাদ
 $wx11$ বাদ
 $wx'10$ বাদ
 $w'z'$ থাকবে শুধু
 wz' থাকবে শুধু
 xz' থাকবে শুধু

SOP form $\rightarrow$ $A+A'=1$ $C+C'=1$
 $B+B'=1$ $D+D'=1$

$$F = A'B'C' + B'CD' + A'BCD' + AB'C'$$

$$= A'B'C'(D+D') + (A+A')B'CD' + A'BCD' + AB'C'(D+D')$$

$$= A'B'C'D + A'B'C'D' + AB'CD' + A'B'CD' + A'BCD' + AB'C'D + AB'C'D'$$



$$F = A'CD' + B'C' + B'D'$$

SOP form

$$F(A, B, C, D) = \sum (0, 1, 2, 5, 8, 9, 10)$$

AB \ CD	C'D'	C'D	CD	CD'
	00	01	11	10
A'B' 00	1	1		1
A'B 01		1		
AB 11				
AB' 10	1	1		1

	8	4	2	1
0 →	0	0	0	0
1 →	0	0	0	1
2 →	0	0	1	0
5 →	0	1	0	1
8 →	1	0	0	0
9 →	1	0	0	1
10 →	1	0	1	0

B'C'

$$F = B'C' + B'D' + A'C'D$$

POS form →

$$F = \sum(0, 1, 2, 5, 8, 9, 10)$$

$$F' = \left(\sum(0, 1, 2, 5, 8, 9, 10) \right)' \Rightarrow F' = \prod(3, 4, 6, 7, 11, 12, 13, 14, 15)$$

$$= \prod(3, 4, 6, 7, 11, 12, 13, 14, 15)$$

$$= M_3 M_4 M_6 M_7 M_{11} M_{12} M_{13} M_{14} M_{15}$$

POS

$$F = \prod(3, 4, 6, 7, 11, 12, 13, 14, 15)$$

POS

AB \ CD	c'd'	c'd	c+d	c+b
A'B'	1 ₀	1 ₁	0 ₃	1 ₂
A'B	0 ₄	1 ₅	0 ₇	0 ₆
A+B	0 ₁₂	0 ₁₃	0 ₁₅	0 ₁₄
A+B'	1 ₈	1 ₉	0 ₁₁	1 ₁₀

c c' वाढ
A A' वाढ
(B+D')

(A+B)

c c' DD' वाढ

c+d
A A', BB' वाढ

$$F = (B+D') \cdot (A+B) \cdot (c+d)$$

09-09-2023

K Map

A \ B	0	1
	$\bar{B}$	B
0 $\bar{A}$		
1 A		

A \ BC	00	01	11	10
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	BC'
0 $\bar{A}$	0	1	3	2
1 A	4	5	7	6

$$F(A, B, C) = \sum (0, 6)$$

$$= \pi (1, 2, 3, 4, 5, 7)$$

AB \ CD	00	01	11	10
	$\bar{C}\bar{D}$	$\bar{C}D$	CD	CD'
00 $\bar{A}\bar{B}$	0	1	3	2
01 $\bar{A}B$	4	5	7	6
11 AB	12	13	15	14
10 AB'	8	9	11	10

$$F(A, B, C) =$$

$$F(x, y, z) = x'y'z' + x'yz' + x'yz + xyz'$$

x \ yz	00	01	11	10
	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
0 x'	1			1
1 x	1			1

$$F(x, y, z) = z'$$

(Ans)

K	2^k box संख्या
0	1
1	2
2	4
3	8
4	16

$$F(w, x, y, z) = \sum (0, 1, 2, 4, 5, 6, 7, 9, 12, 13, 14)$$

	$w'z'$	$w'z$	wz	wz'	
$w'x'$	1	0	1	1	3
$w'x$	1	4	1	5	7
wx	1	12	1	13	15
wx'	1	8	1	9	11

$w'z'$ বাদ
 xx' বাদ
 wz বাদ
 $w'z'$ বাদ

$$F = w'z' + y' + xz'$$

Total 8 টি 1 আছে
 $2^3 = 8$

$4 - 3 = 1$ → power
 variable

একটি চমক থাকবে কুণ্ডু

(০০০)

$x = (\sum w'x)7$

AB \ CDE								
	C'D'E'	C'D'E	C'DE	C'DE'	CDE'	CDE	CD'E	CD'E'
	000	001	011	010	110	111	101	100
A'B' 00	0	1	3	2	6	7	5	4
A'B 01	8	9	11	10	14	15	13	12
AB 11	24	25	27	26	30	31	29	28
AB' 10	16	17	19	18	22	23	21	20

$$F(A, B, C, D, E) = \sum (0, 2, 4, 6, 9, 11, 13, 15, 17, 21, 25, 27, 29, 31)$$

A'B' \ CDE								
	C'D'E'	C'D'E	C'DE	C'DE'	CDE'	CDE	CD'E	CD'E'
A'B'	1			1	1			1
A'B		1	1			1	1	
AB		1	1			1	1	
AB'		1					1	

BB'
CC'

AD'E

AA' DD' CC'
DE

CC' DD'
A'B'E

ABD 000 001 011 010 110 111 101 100
 001 010 011 100 101 110 111 000

{ 000
 001
 011
 010
 110
 { 111
 101
 100

C
 B
 A

000	001	011	010	110	111	101	100
	*					*	
	#					#	
	#					#	
	*					*	

ABD

BD
CD

POS

$$F(x, y, z) = \pi(0, 2, 5, 7)$$

	$y'z$	yz'	yz	$y'z'$
$x=0$	0	1	3	2
$x=1$	4	5	7	6

$$F = (xz + x'z') = \overline{xz} \cdot \overline{x'z'} = (x' + z') \cdot (x + z)$$

SOP

$$F(x, y, z) = \sum(2, 3, 6, 7)$$

	$y'z'$	$y'z$	yz	yz'
$x=0$			1	1
$x=1$			1	1

$$F = y$$

$$F(A, B, C, D) = \sum(7, 13, 14, 15)$$

	$C'D'$	$C'D$	CD	CD'
$A'B'$				
$A'B$		X	1	
AB	1	1	1	1
AB'				

$$A A'$$

$$AB C$$

$$D D'$$

$$= ABC + BCD + ABD$$

$$= AB(C + D) + BCD$$

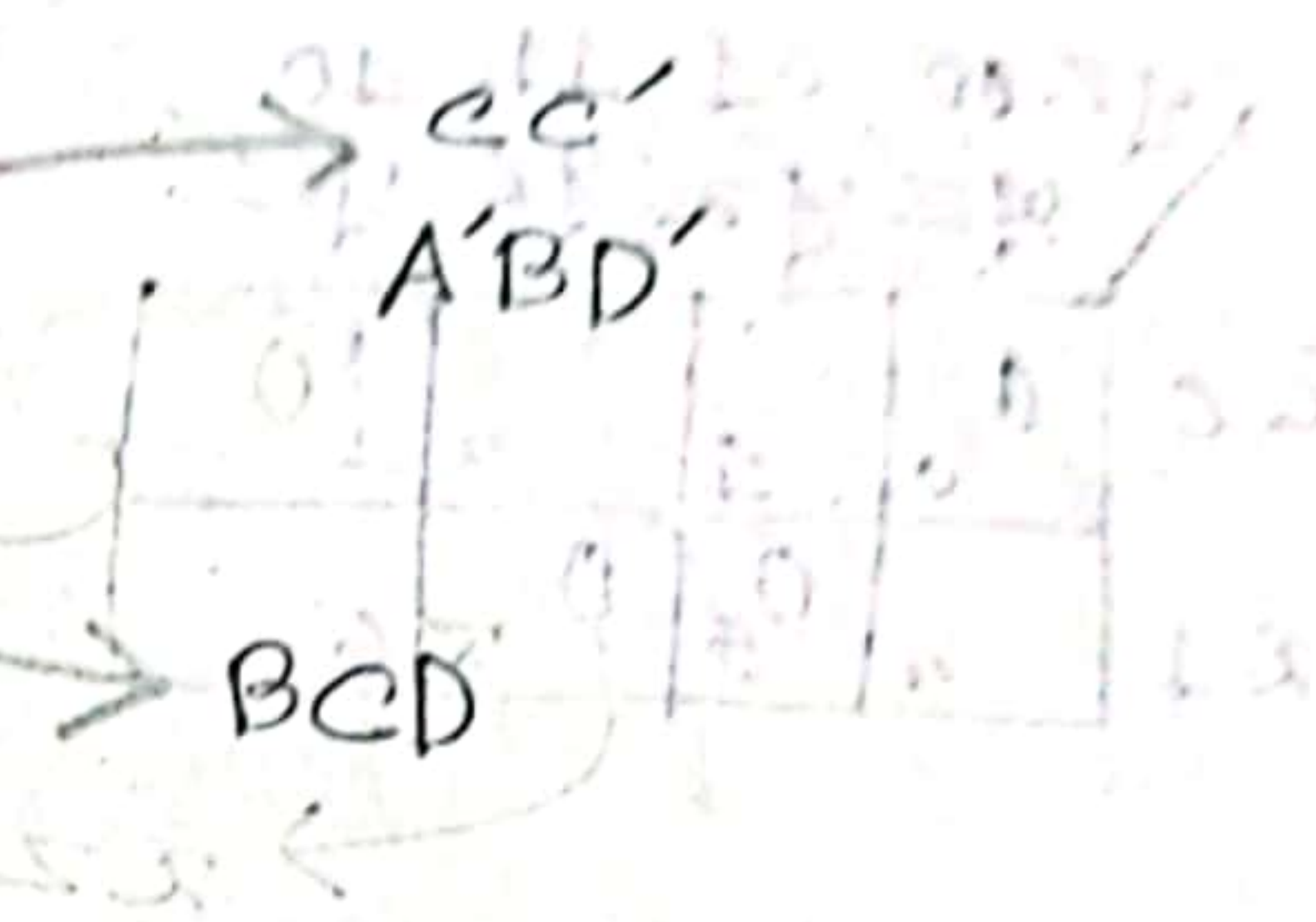
$$F =$$

$$CC'$$

$$ABD$$

$$F(A, B, C, D) = \sum (4, 6, 7, 15)$$

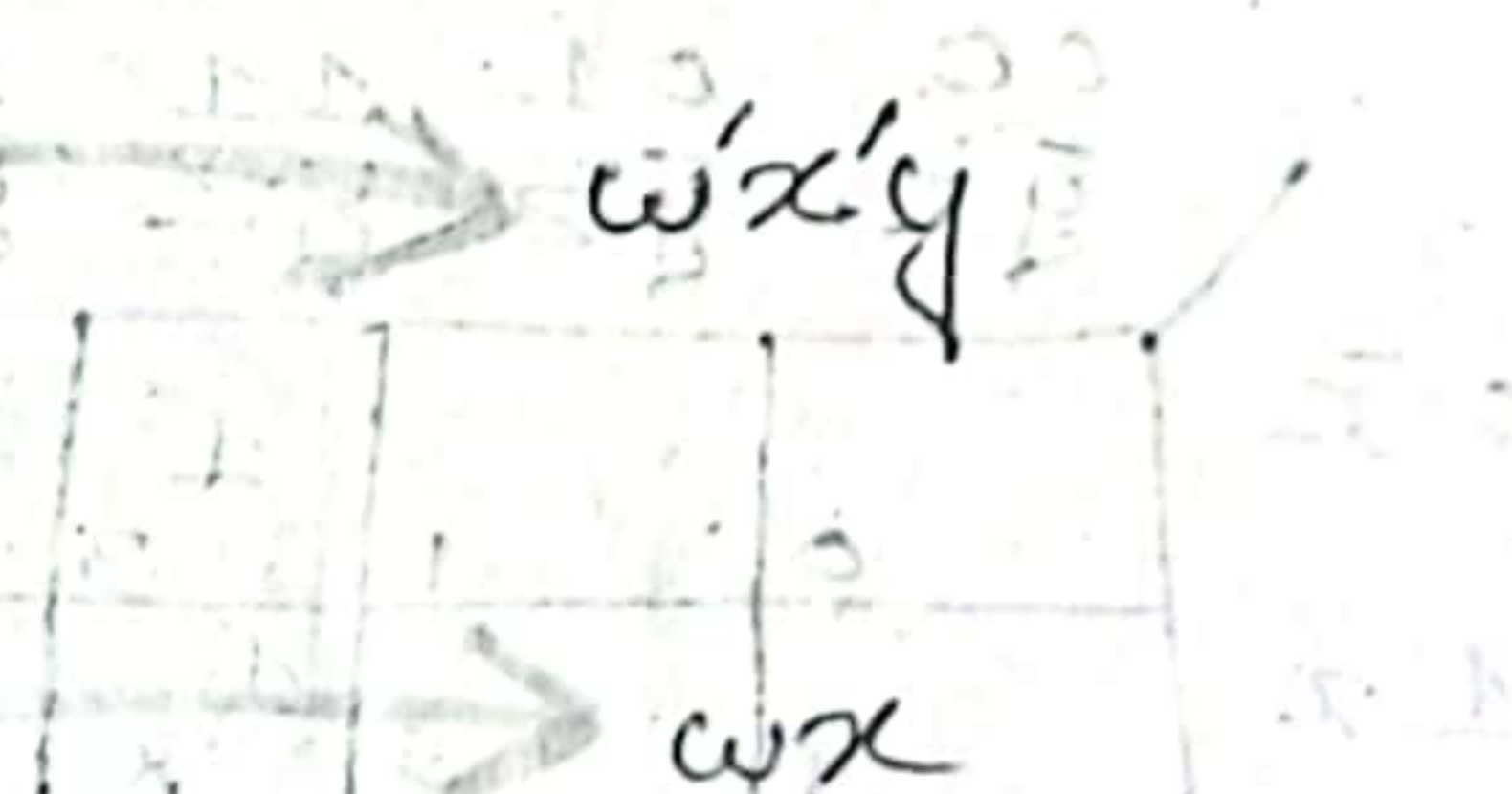
AB \ CD	C'D'	C'D	CD	CD'
A'B'				
A'B	1		1	1
AB			1	
AB'				



$$F = A'BD' + BCD$$

$$F(w, x, y, z) = \sum (2, 3, 12, 13, 14, 15)$$

wx \ yz	y'z'	y'z	yz	yz'
w'x'			1	1
w'x				
wx	1	1	1	1
wx'				



$$F = wx'y + wx$$

POS

$$F(x, y, z) = \prod (0, 1, 4, 5)$$

V.V.I.

যদি SOP term

যেখানে বারি আছে x

যা result আমলে x

অর আবার complement

করতে হবে। (পুরোটা part)

$$F = (x')'$$

$$= x \text{ Ans}$$

SOP

$yz \quad yz' \quad y'z \quad y'z'$

0	0		
0	0		

ডি-মরগানের সূত্রে
Simplify করতে then
result আমলে

POS $\rightarrow$ low signal

yz	$y+z$ 00	$y+z'$ 01	$y'+z'$ 11	$y'+z$ 10
x	0	0	1	3
x'	0	0	5	7

x, x' বাদ
 z, z' বাদ
 y

$$F = y \text{ (Ans)}$$

$$F(A, B, C, D) = \prod (0, 1, 2, 3, 4, 10, 11)$$

AB	$C'D'$	$C'D$	CD	CD'
$A'B'$	0	0	0	0
$A'B$	0			
AB				
AB'			0	0

$(A'C'D')$

$$F = (A'C'D') + (A'B) + (CB')$$

$$= (A+C+D) \cdot (A+B') \cdot (C'+B)$$

POS

$$F(x, y, z) = \pi(1, 3, 5, 7, 13, 15)$$

	$y'z'$	$y'z$	$y \cdot z$	$y \cdot z'$	
ωx					$\omega'z$
$\omega'x'$	0	1	0		0
ω	4	5	7	6	
$\omega \cdot x$	12	13	15	14	
$\omega'x'$	8	9	11	10	

$$F = (\omega'z) \cdot (x+z)$$

$$(x+y) \cdot (x+z) + (x+y) \cdot (x+z) = 1$$

$$(x+y) \cdot (x+z) + (x+y) \cdot (x+z) = 1$$

Don't Care → একে 0 বা 1 উভয় হিসাবে ব্যবহার করা যায়।
 X দিয়ে প্রকাশ করা হয়

$F(w,x,y,z) = \sum (1, 3, 7, 11, 15)$

$D(w,x,y,z) = \sum (0, 2, 5)$

→ Don't care

অসংখ্য don't care-কে ধাক্কা এর আওতায় আনাতে হবে এমন কোনো বাধ্যবাধকতা নেই।

$w \backslash yz$	$y'z'$	$y'z$	yz	yz'
$w'x'$	X 0	1 1	1 3	X 2
$w'x$	4	X 5	1 7	6
wx	12	13	1 15	14
wx'	8	9	1 11	10

→ $w'x'$
 → yz $BA + CA + DA = 7$

$F = \sum (1, 3, 4, 7, 13)$

$D = \sum (11, 14, 15)$

→ Don't care

$AB \backslash CD$	$C'D'$	$C'D$	CD	CD'
$A'B'$		1 1	1 3	2
$A'B$	1 4	5	1 7	6
AB	12	1 13	X 15	X 14
AB'	8	9	X 11	10

CC' বাদ
 $A'B'D$

AA', BB' বাদ
 CD

ABD
 CC' বাদ

$F = CD + A'B'D + A'BC'D' + ABD$

$$F = \sum m(0, 2, 3, 6, 7, 12, 13, 14) + \sum d(1, 4, 11, 15)$$

don't care ←

AB \ CD	c'd'	c'd	cd	cd'
00 A'B'	1	X	1	1
01 A'B	X		1	1
11 AB	1	1	X	1
10 AB'			X	

00 A'B' → DD' BB' and A'C
 11 AB → CC' DD' and AB

$$F = A'B' + A'C + AB$$

AB \ CD	c'd'	c'd	cd	cd'
00 A'B'	1		1	1
01 A'B			1	1
11 AB	1	1		1
10 AB'			1	

$$F = CD + A'B'D + A'BCD + ABD$$

POS

$$F = \pi M(5, 8, 9, 10) \cdot \pi D(1, 4, 11, 15)$$

Don't care

Maxterm

$$0 = A$$

$$1 = A'$$

Minterm

$$0 = A'$$

$$1 = A$$

AB \ CD	00	01	11	10
00		X		
01	X	0		
11			X	
10	0	0	X	0

DD' = 1

$$(A' + B + C')$$

$$(A' + B)$$

$$F = (A' + B + C') \cdot (A' + B)$$

POS

L (Loco signal) নিয়ে কাজ করে

0 → A
1 → A'

EOP

0 → A'
1 → A

		B	
		0	1
A	0	$A+B$ 0	$A+B'$ 1
	1	$A'+B$ 2	$A'+B'$ 3

		BC			
		00	01	11	10
		$B+C$	$B+C'$	$B'+C'$	$B'+C$
A	0	$A+B+C$ 0	$A+B+C'$ 1	$A+B'+C'$ 3	$A+B'+C$ 2
	1	$A'+B+C$ 4	$A'+B+C'$ 5	$A'+B'+C'$ 7	$A'+B'+C$ 6

		CD			
		00	01	11	10
		$C+D$	$C+D'$	$C'+D'$	$C'+D$
AB	00	$A+B+C+D$ 0	$A+B+C+D'$ 1	$A+B+C'+D'$ 3	$A+B+C'+D$ 2
	01	$A+B'+C+D$ 4	$A+B'+C+D'$ 5	$A+B'+C'+D$ 7	$A+B'+C'+D'$ 6
	11	$A'+B'+C+D$ 12	$A'+B'+C+D'$ 13	$A'+B'+C'+D'$ 15	$A'+B'+C'+D$ 14
	10	$A'+B+C+D$ 8	$A'+B+C+D'$ 9	$A'+B+C'+D'$ 11	$A'+B+C'+D$ 10

□ $F = (A+B) \cdot (A+B)$ যা $F = \sum (0, 2)$

	B 0	B 1
A 0	0	0
A' 1	0	0

→ AA' বাদ
B

$$\begin{aligned} & A'A + AB + A'B + BB \\ &= AB + A'B + B \\ &= B(A+A') + B \\ &= B + B \\ &= B \\ & \text{(Proved)} \end{aligned}$$

$F = B$

□ $F = (A+B+c') \cdot (A+B'+c') \cdot (A+B+c)$

	BC 00	B+c' 01	B'+c' 11	B'+c 10
A 0	0	0	0	$A+B+c$
A' 1	$A+B+c$	$A+B+c'$	$A+B'+c'$	$A+B'+c$

cc' বাদ
(A+B)

BB' বাদ
(A+c')

$F = (A+B) \cdot (A+c')$

□ $F = \sum (6, 7, 15)$

$F = (A+B'+c') \cdot (B'+c'+D')$

	CD 00	c+D' 01	c'+D' 11	c+D 10
A+B 00	0	1	3	2
A+B' 01			0	0
A'+B' 11			0	
A'+B 10				

AA' বাদ
(B'+c'+D')

DD' বাদ
 $A+B'+c'$